

Week 12
Planning algorithms

ENN581 Robot Motion, Control and Planning

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ACKNOWLEDGEMENT OF TRADITIONAL OWNERS

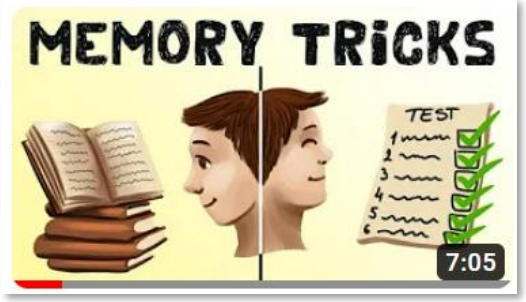
QUT acknowledges the Turrbal and Yugara, as the First Nations owners of the lands where QUT now stands. We pay respect to their Elders, lores, customs and creation spirits. We recognise that these lands have always been places of teaching, research and learning.

QUT acknowledges the important role Aboriginal and Torres Strait Islander people play within the QUT community.

Exam Prep Tips: Week 12

Create mnemonics to memorise formulas

- Chunking information, acronyms and acrostics can improve your ability to recall formulas (remember All-Silver-Tea-Cups from high-school maths?).
- Watch **this video** (7 min.) to see example mnemonics.



Try the Pomodoro Technique to beat procrastination

- Short intervals are less daunting and, therefore, more productive.
- 25 min. study time + 5 min. break (repeat)
- Use a timer to keep you accountable.
- Watch **this video** (2 min.) for details.



See the full list of tips in the **Guide to Succeeding in Engineering Exams**

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So far, we have learnt

Forward and inverse kinematics – manipulators and mobile robots

Forward and inverse dynamics - manipulators and mobile robots

Trajectory generation

Choosing appropriate control inputs

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What does planning mean?

- Different things to different people.
- Here, we focus on motion planning / path planning for robotics.
- Given a map, a start location and a goal location, identify a trajectory that the robot should take to reach the goal when executed.
- We are also interested in considering some of the constraints of the robot, ensuring that our generated plans are feasible.

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Today's lecture

- Configuration space transform
- Path planning algorithms
  - Representing the planning problem as graphs
  - Searching the graphs for a solution
  - Grid-based vs sampling-based approaches to path planning

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Path planning

The goal:

- Compute a collision-free path for a rigid or articulated object (the robot) among static obstacles.

Inputs:

- Geometry of robot and obstacles (map)
- Kinematics of robot (degrees of freedom, constraints)
- Initial and goal robot configurations (poses)

Outputs:

- A continuous sequence of collision-free robot configurations connecting the initial and goal configurations
- Report failure if no such path exists.

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Simplifying assumptions

- Our representation of the robot and the world is sufficiently expressive.
- We know where we are and where we want to go.
- We have a motion model for our robot.

- The path planning problem is typically cast as an optimization problem
  - Minimize time or minimize distance etc. within constraints.

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Trends in motion planning

Classical Robotics (mid-70s)
- Exact models
- No sensing necessary

Reactive Paradigm (mid-80s)
- No models
- Heavily relied on sensing

Hybrid (90s)
- Used models at a high-level
- Reactive at lower level

Probabilistic models (since mid-90s)
- Integrated models and sensing
- Inaccurate models and sensors

Based on G Hager

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Representation

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Representation is important

- How the world is represented and understood by the planner is important.
- There is usually some degree of simplification in choosing a representation.
- By choosing a suitable representation, we may be able to apply existing algorithms to solve our planning problem.

What did we learn about representations so far?

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Task space

The frame of reference in which a robot performs its task.

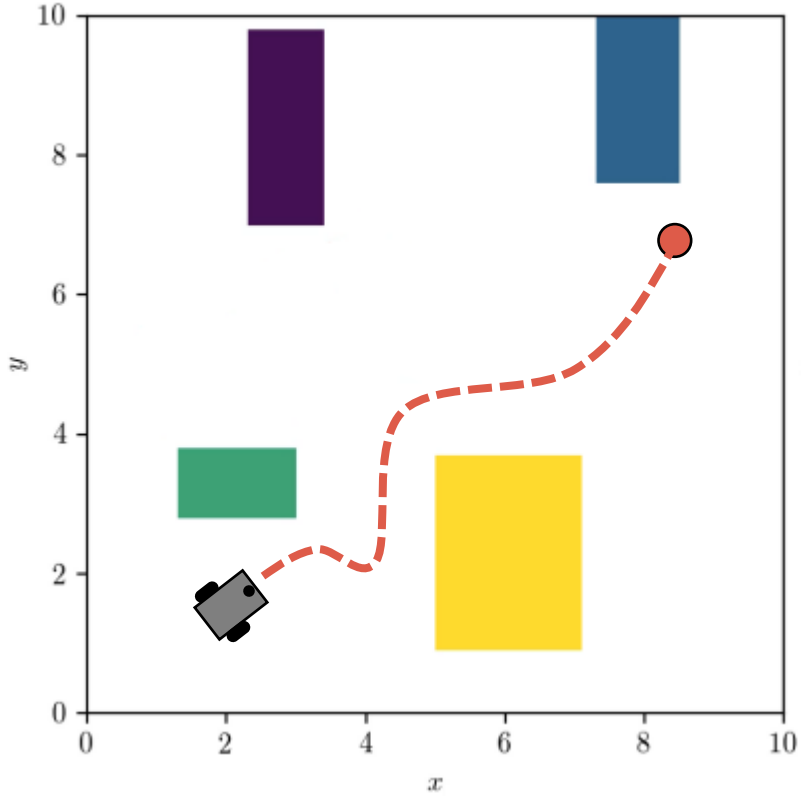
(can be independent of the description of the robot)

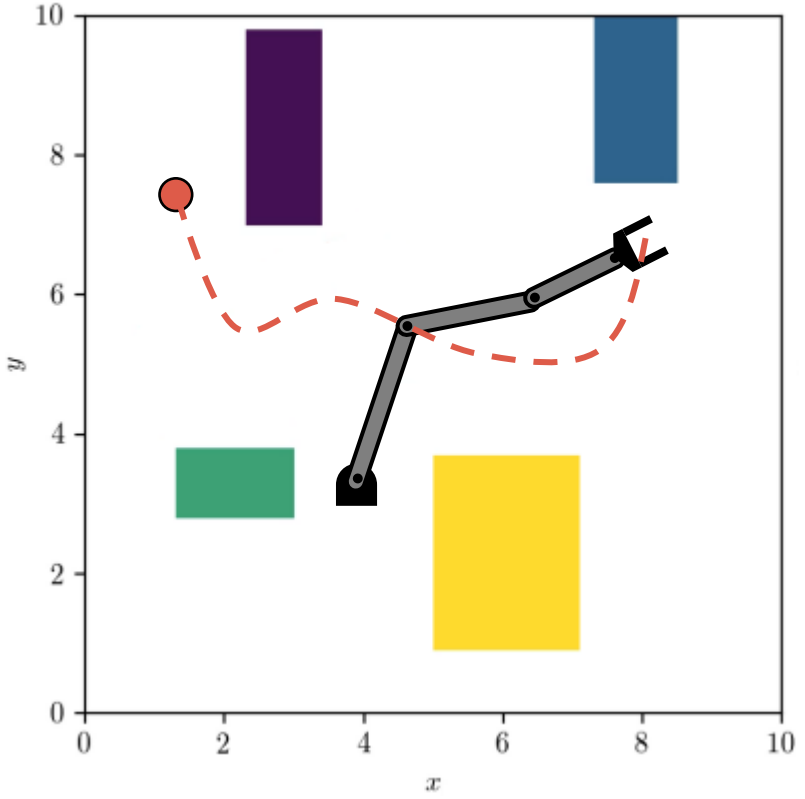
Examples:

- a reference frame attached to the end effector
- a frame attached to a mobile robot.

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Task space





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Configuration space

The set of generalised coordinates which completely describes the configuration of a robot

- Joint parameters for a manipulator robot
- Position and orientation of a mobile robot

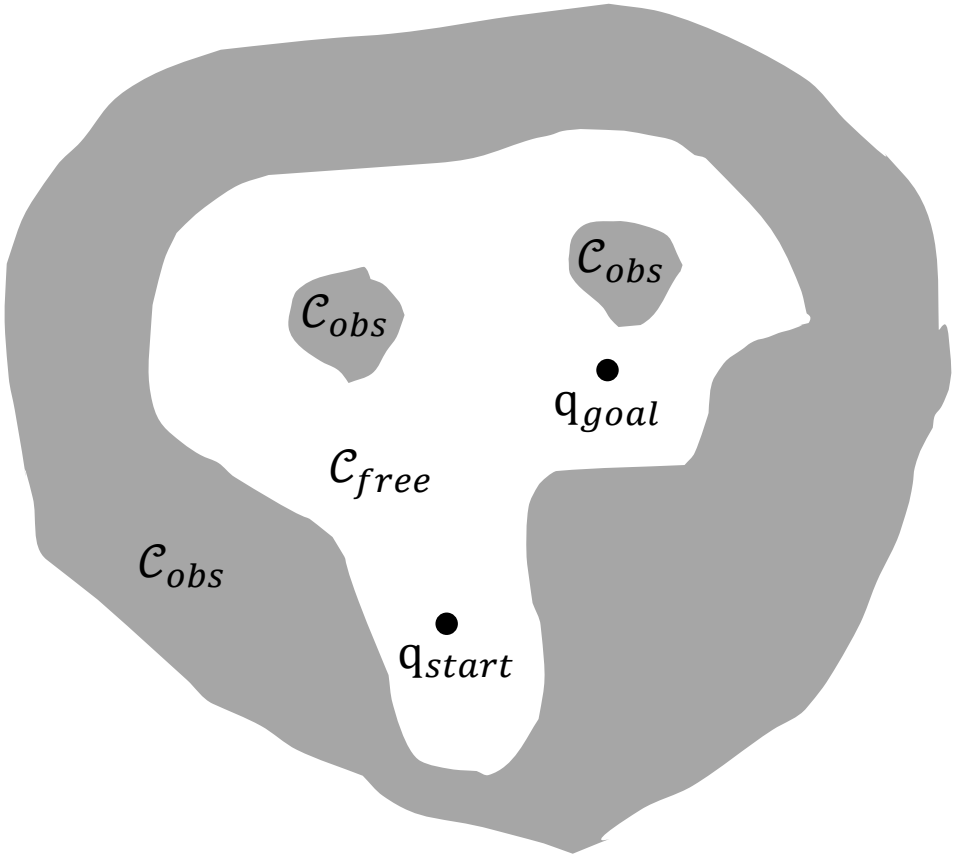
It is common to do path planning in the configuration space of the robot.

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Free space and obstacle region

The configuration space can be thought of as a combination of free space and that occupied by obstacles

- Let $\mathcal{C}$ be the configuration space and $\mathcal{W}$ be the workspace of the robot.
- Obstacles occupy a portion of the workspace. $\mathcal{O} \in \mathcal{W}$ or $\mathcal{C}_{obs} \in \mathcal{C}$
- The robot's state is a point in the configuration space. $\mathrm{q} \in \mathcal{C}$
- Free space is all the possible points in the configuration space that are not occupied by obstacles. $\mathcal{C}_{free} = \{\mathrm{q} \in \mathcal{C} \mid \mathrm{q} \cap \mathcal{C}_{obs} = \phi\}$

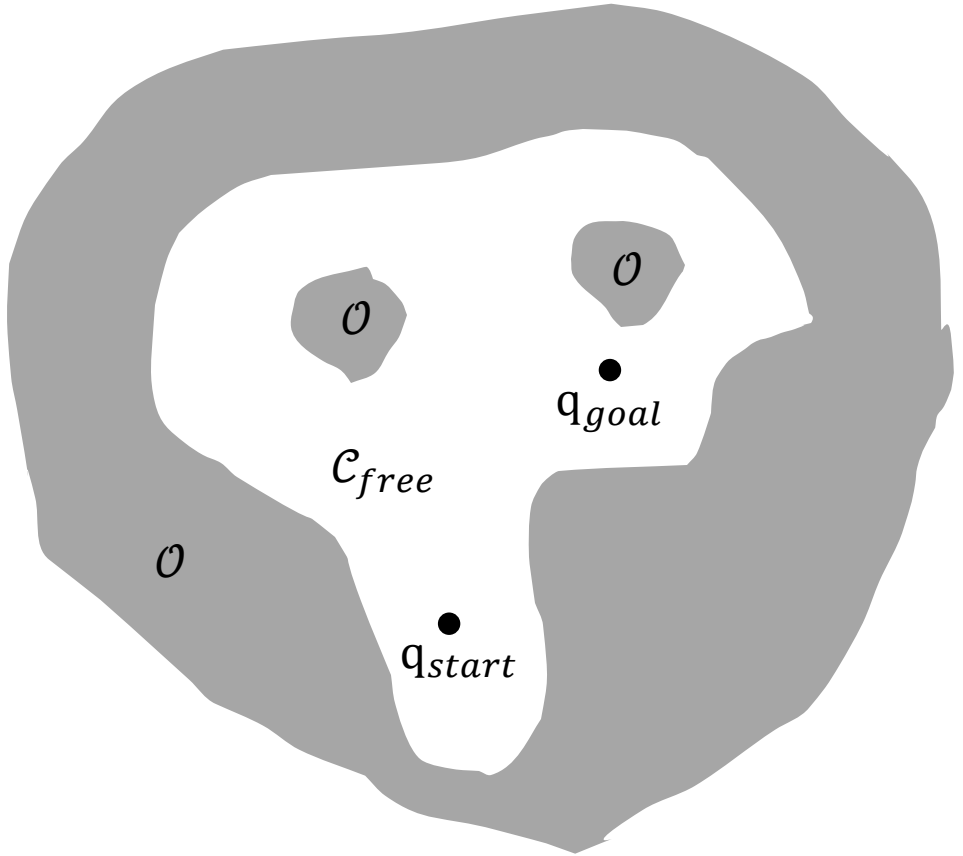


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Path planning is identifying a set of points for the robot to follow in the free space

- In this setting, we can plan with the robot begin a point in the free region of the configuration space.

  A set of q in $\mathcal{C}_{free}$



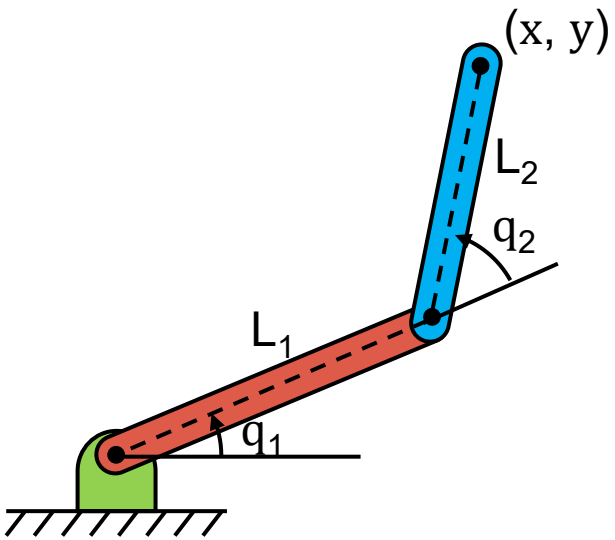
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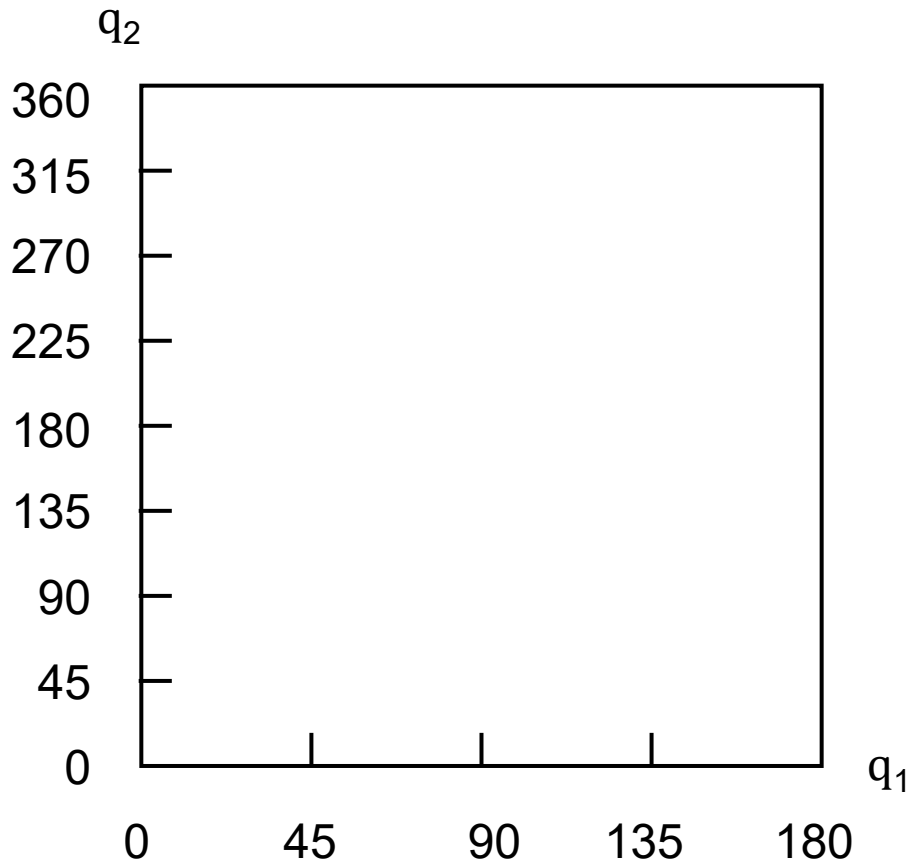
Why use configuration space?

- Positions in configuration space tend be close together for the robot.
- Can be easier to solve collision checks and join nearby poses.
- Allows a level of abstraction that means solution methods can solve a wider range of problems.
- Sometimes helps with wraparound conditions
  (remember the case with rotation joints in the assessment?)

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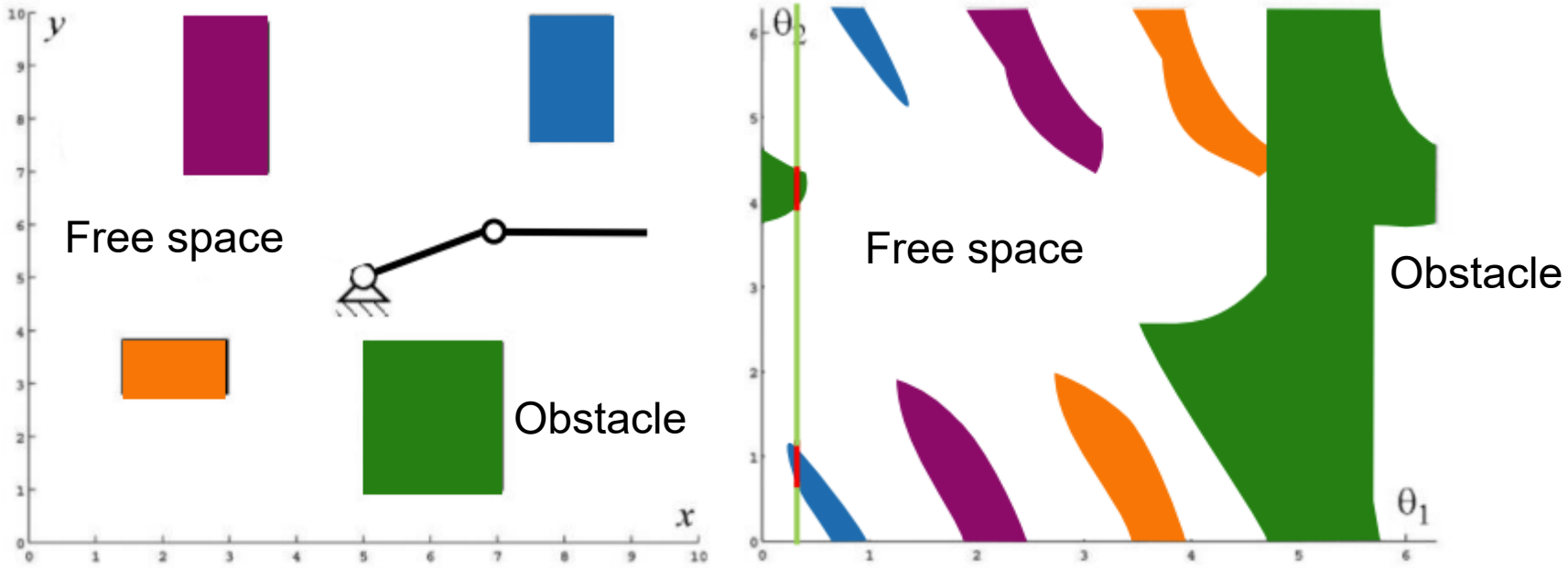
Example: for a manipulator arm





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Example: for a manipulator arm

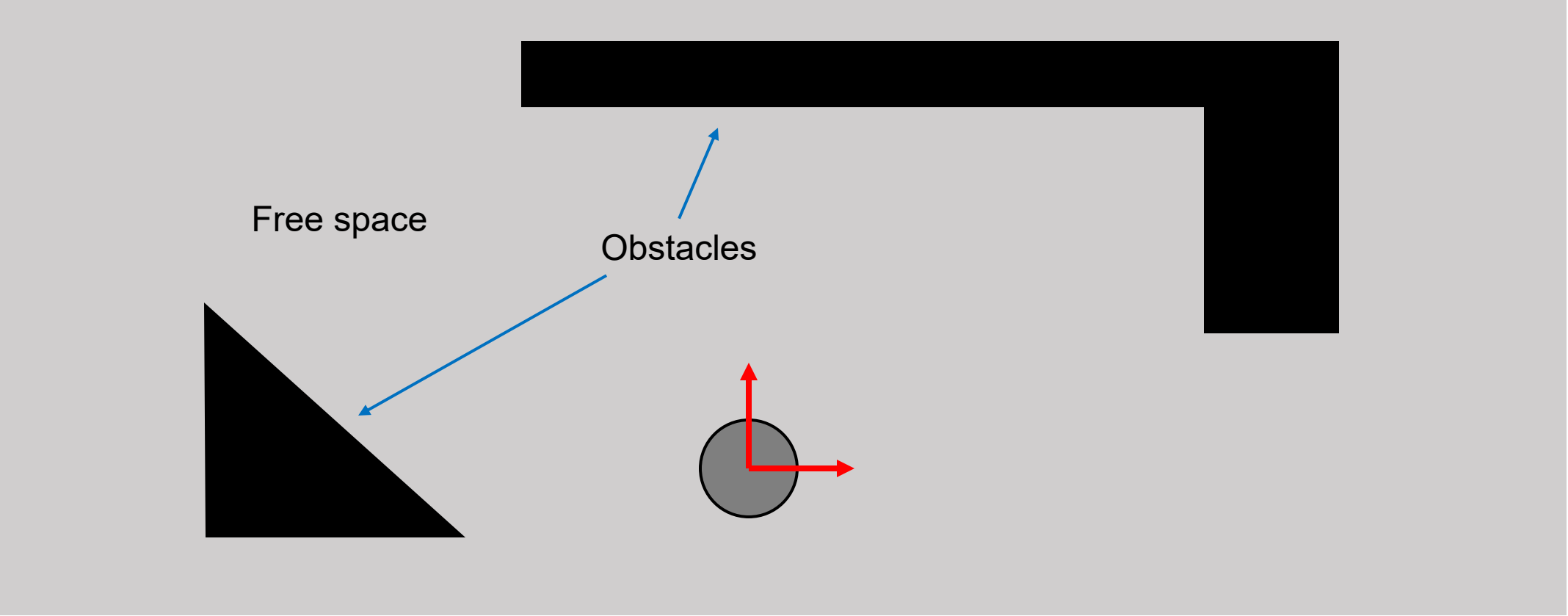


Task space

Configuration space

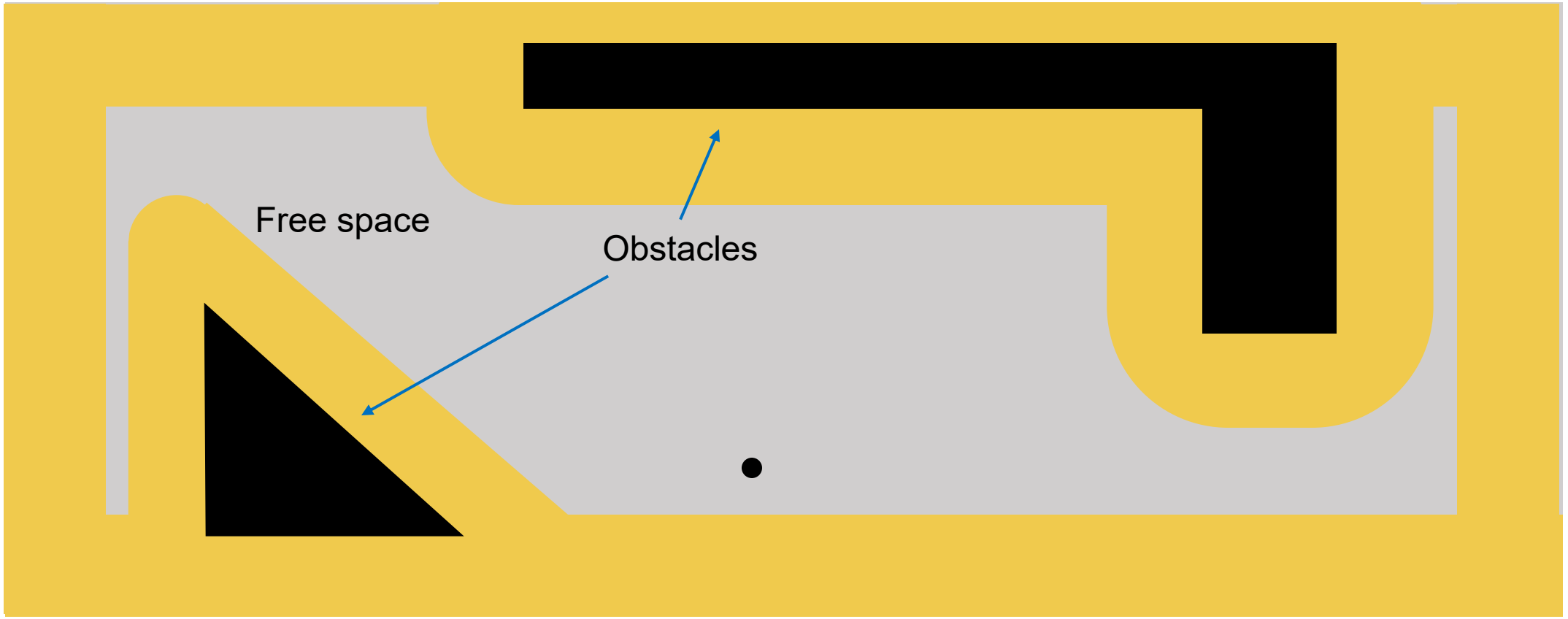
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Example: for a mobile robot



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The robot is reduced to a point and the obstacles inflated

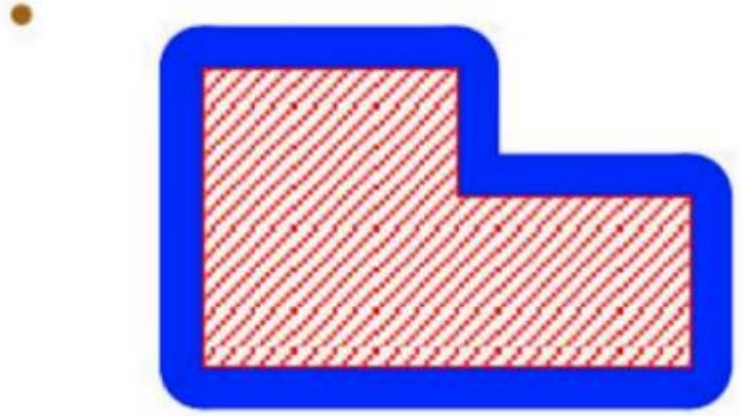


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The robot is reduced to a point and the obstacles inflated



Where can the robot move in the vicinity of the obstacle?



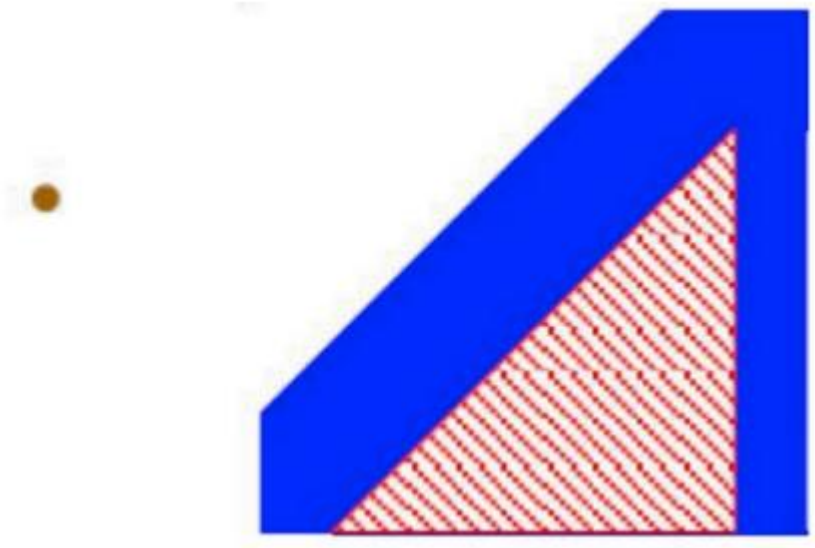
Where can the point move in the vicinity of the inflated obstacle?

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The robot is reduced to a point and the obstacles inflated



Where can the robot move in the vicinity of the obstacle?



Where can the point move in the vicinity of the inflated obstacle?

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Configuration space transformation

- Useful for path planning
- We turned the problem from

  “Twist and turn this 2D polygon past this other 2D polygon”

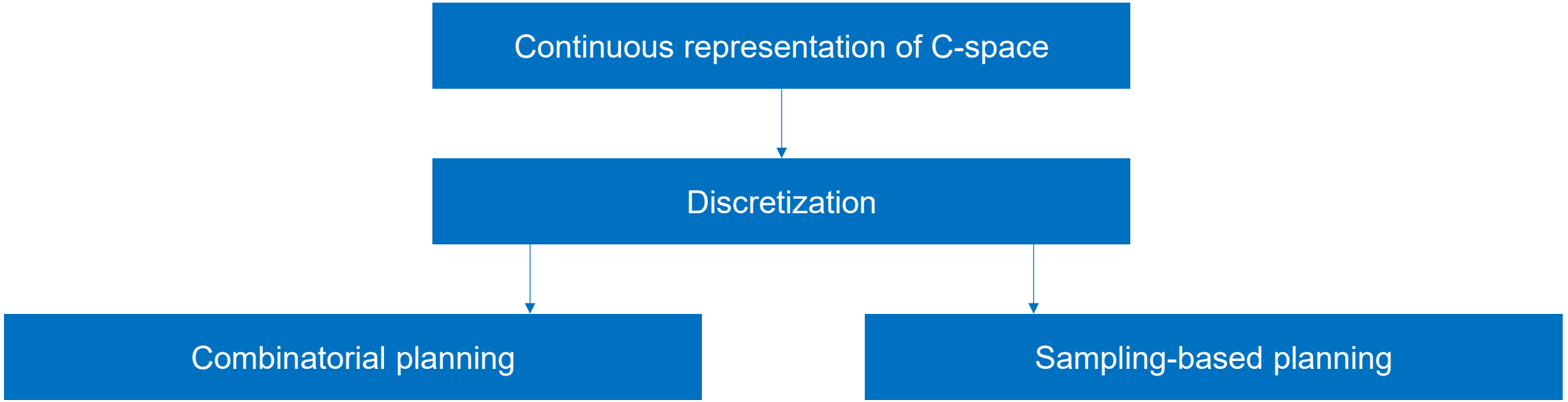
  into

  “Find a path for this point in 2D space past this augmented 2D polygon”

- Now, we can plan paths for points instead of polyhedrons even in the case of mobile robots.

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Framework for path planning



Characterizes $\mathcal{C}_{free}$ explicitly by capturing the connectivity of $\mathcal{C}_{free}$ into a graph and finds solutions using search

Uses collision-detection to probe and incrementally search the C-space for a solution

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Why discretise?

- Although continuous representations have nice mathematical properties, they aren't always convenient.
- Since computers store things digitally, there are some clever and efficient ways to create (approximate) discrete representations
- It is very common to convert a planning problem to some kind of (discrete) graph representation, then use one of a variety of existing search algorithms on the graph

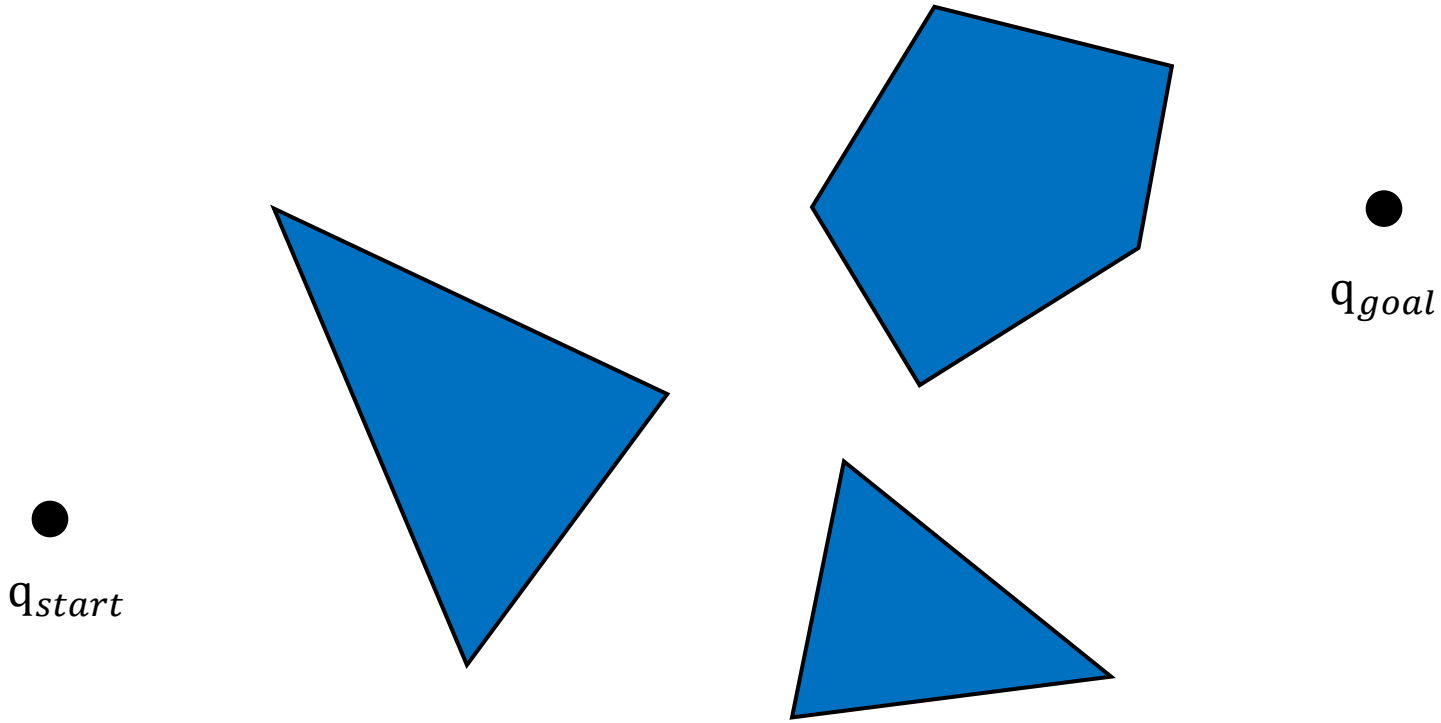
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Visibility graphs

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Visibility graphs

Consider the following configuration space with polygonal obstacles.

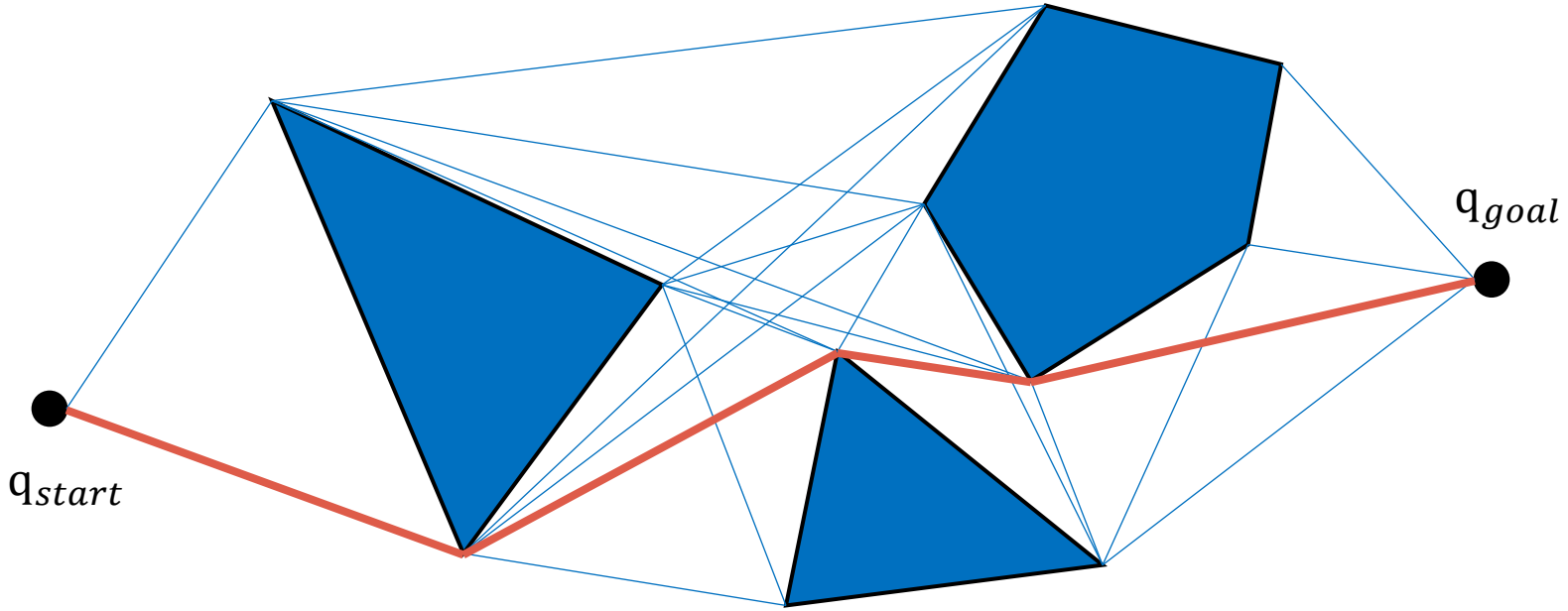


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Visibility graphs

Find all non-blocked lines between polygon vertices, start and goal.

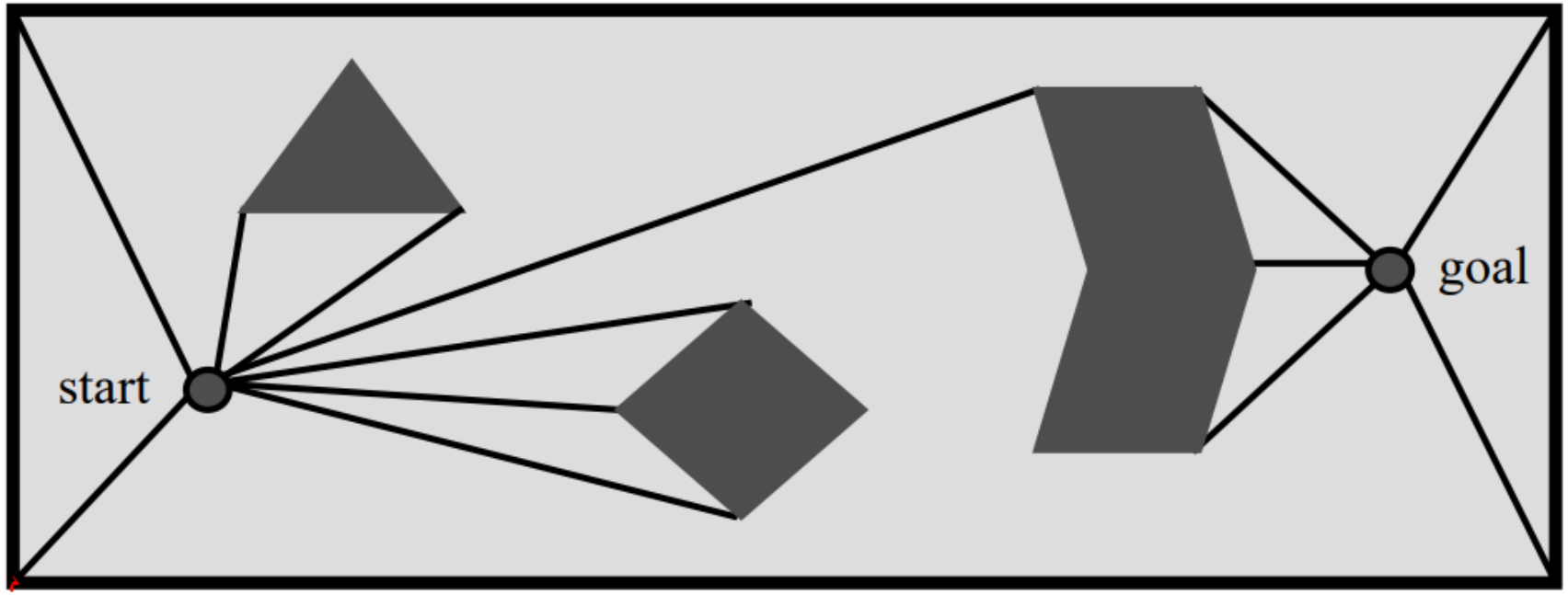
Search the graph of these lines for the shortest path



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Visibility graph algorithm

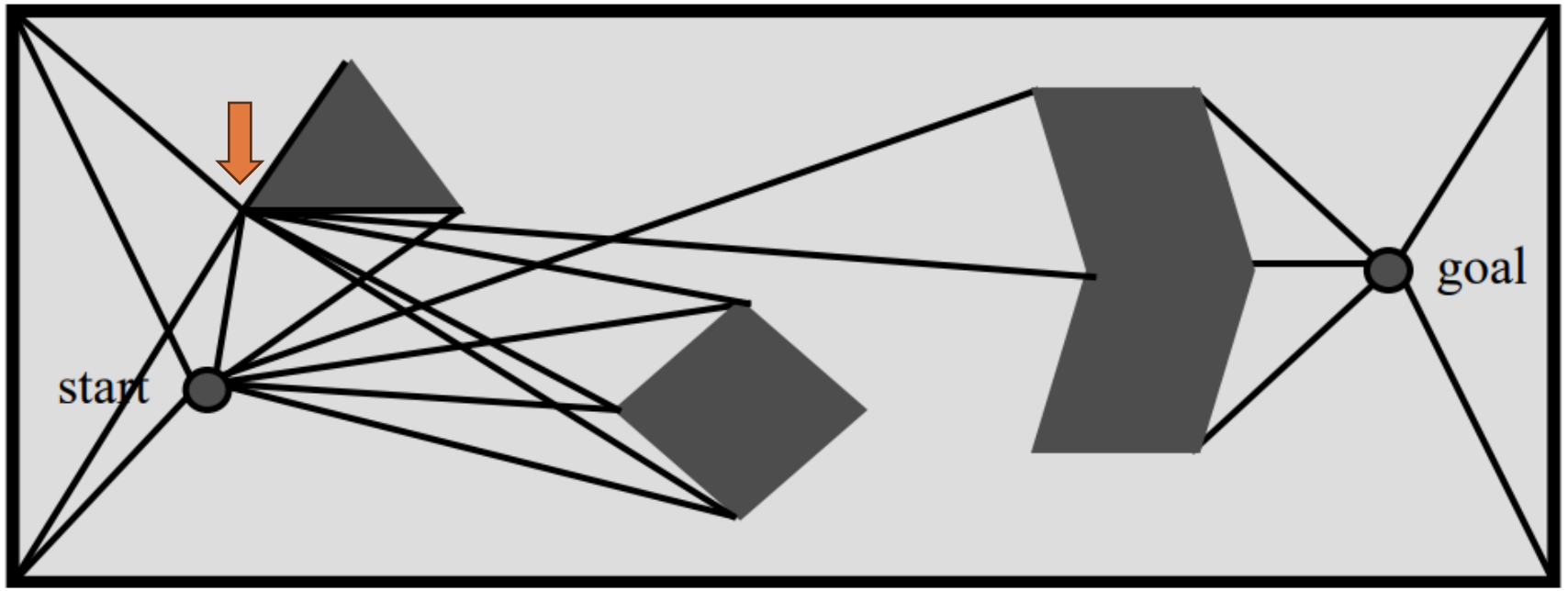
First, draw lines from the start and the goal to all the visible vertices and corners of the world



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Visibility graph algorithm

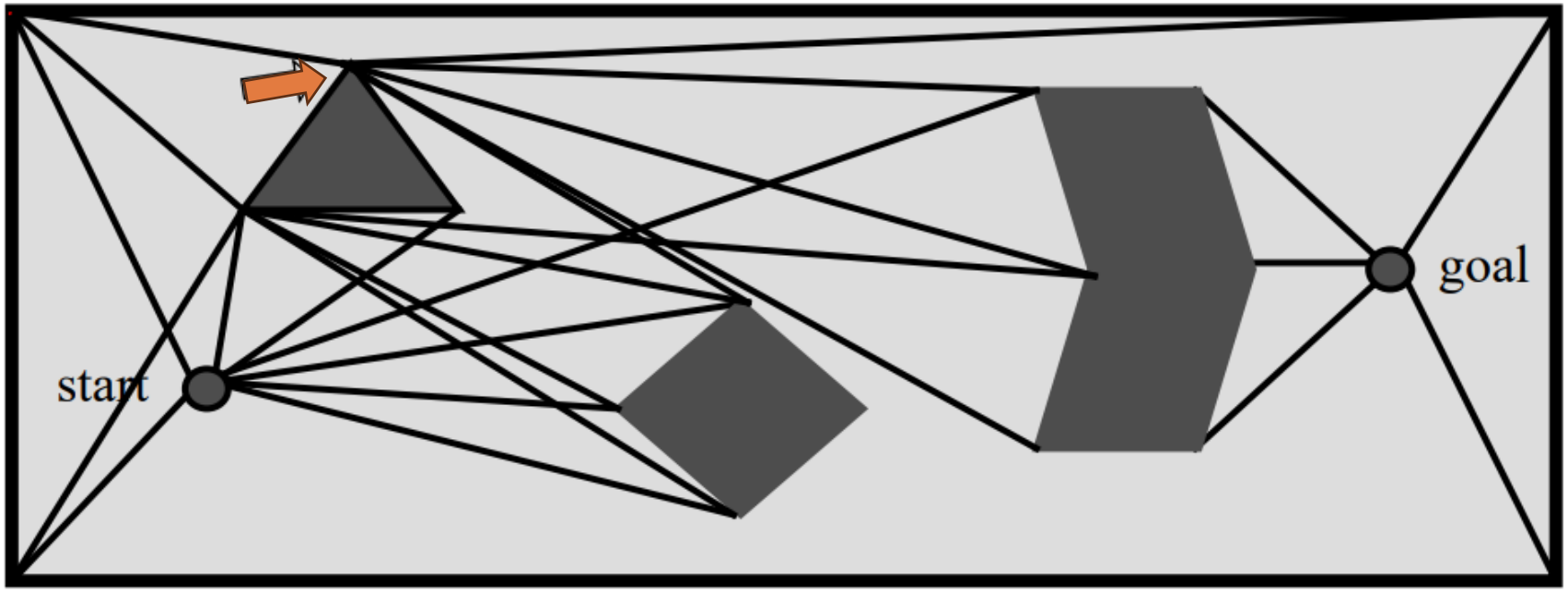
Next, draw lines of sight from every vertex of every obstacle like before. Remember that lines along edges are also lines of sight.



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Visibility graph algorithm

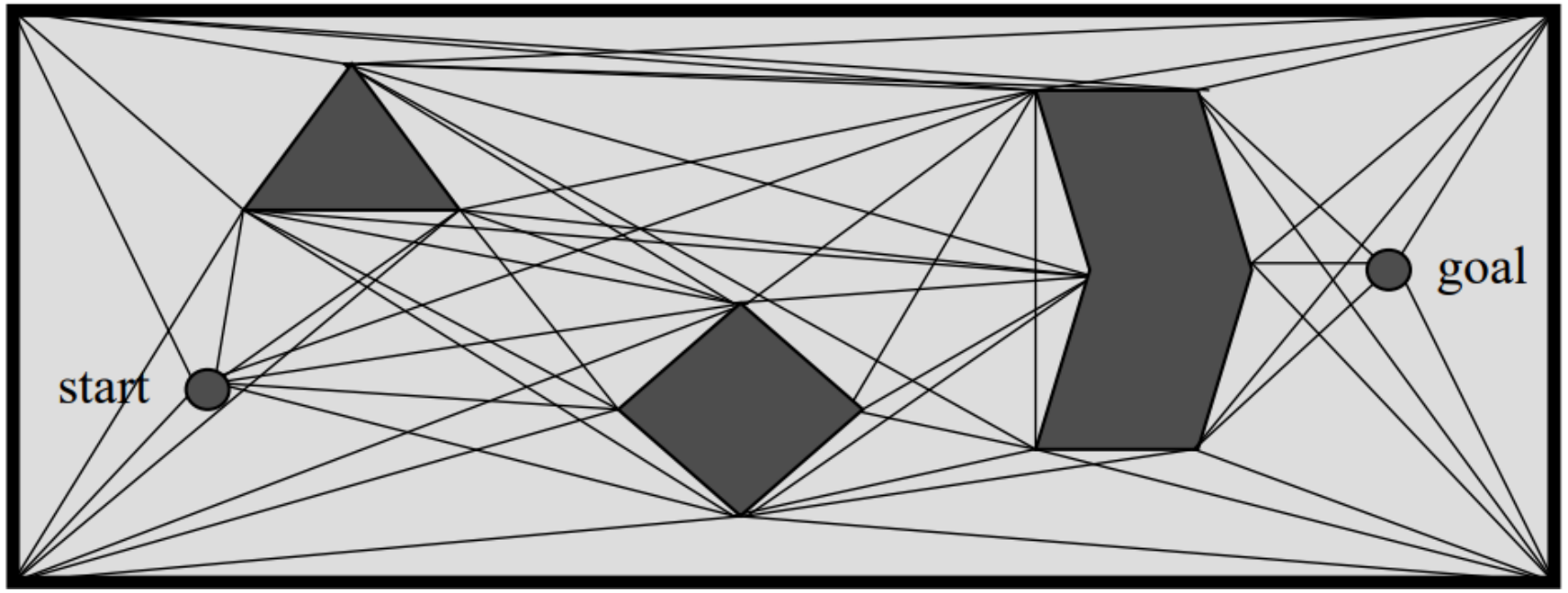
Continue drawing lines of sight from each vertex



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Visibility graph algorithm

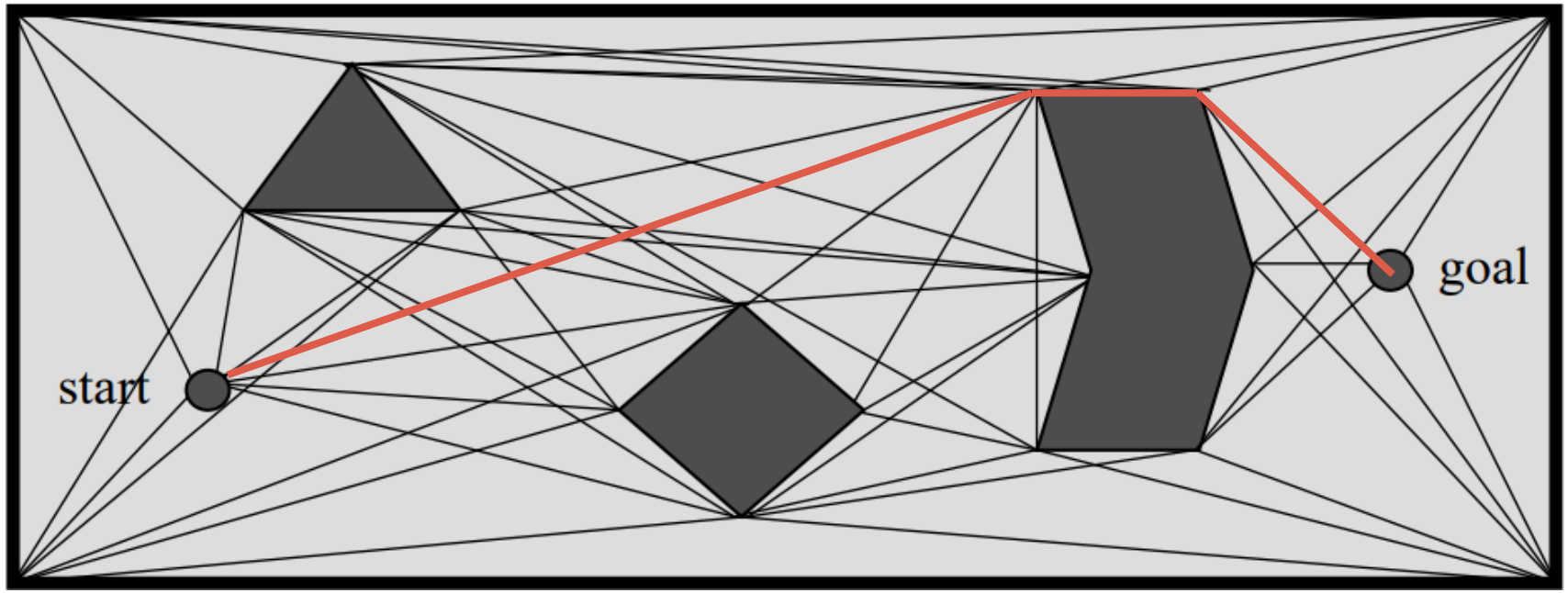
Repeat until done



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Visibility graph algorithm

Search the graph for the shortest path



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Visibility graph – pros and cons

- Pros
  - It is easy to find the shortest path from the start to goal positions.
  - Implementation is simple in the case of polygon obstacles.

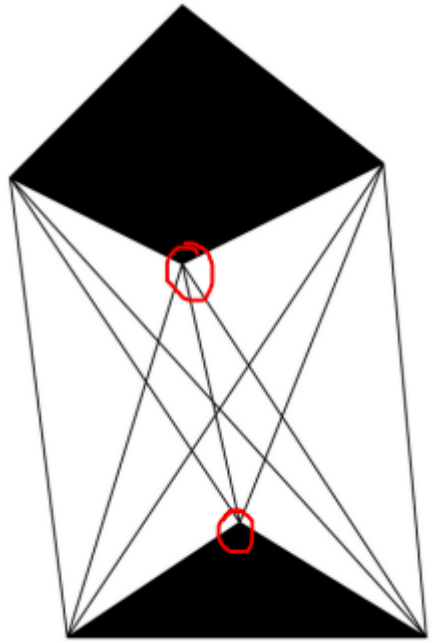
- Cons
  - Number of nodes and edges increases with the number of polygons.
  - Thus, it can be inefficient in densely populated problems.
  - The solution path takes the robot as close to the obstacle as possible.

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Reduced visibility graph

The lines that are useful for planning a path are the ones

- that go around the object,
- or those that pass through the extremes of the object.



We can ignore the lines that pass through the circled points in certain cases

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Voronoi diagrams

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Voronoi diagrams

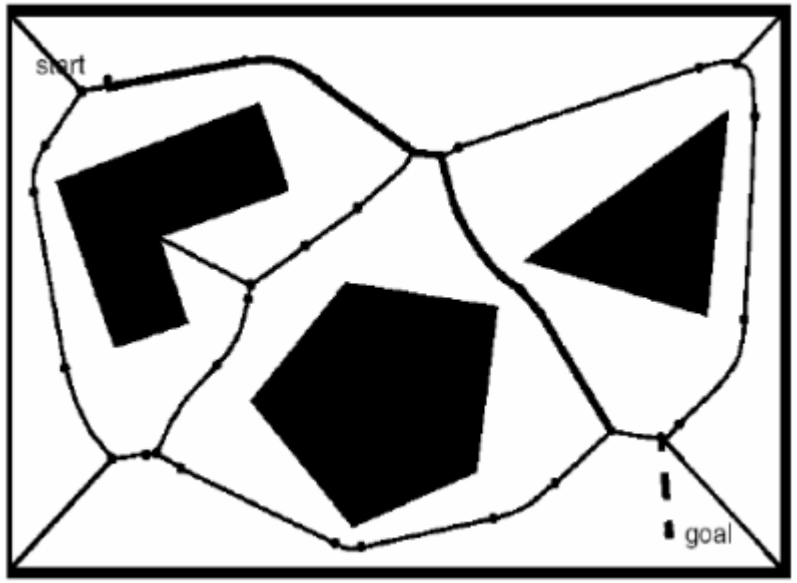
Key idea:

- We don’t care about the optimum path
- We want to get a collision-free path as far away from obstacles as possible

- In contrast to visibility graphs, a Voronoi graph tends to maximise the closest distance to obstacles.
- Easy to implement: maximise sensor readings

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Voronoi diagram



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Voronoi diagram – pros and cons

Pros:

- Using range sensors like a sonar or laser scanner, a robot can navigate along the Voronoi diagram using simple control rules

Cons:

- Because the Voronoi diagram tends to keep the robot as far away from the obstacle as possible, a short-range distance sensor will be in danger of failing.

Peculiarities:

- when the obstacles are polygons, the Voronoi diagram consists of straight and parabolic segments.

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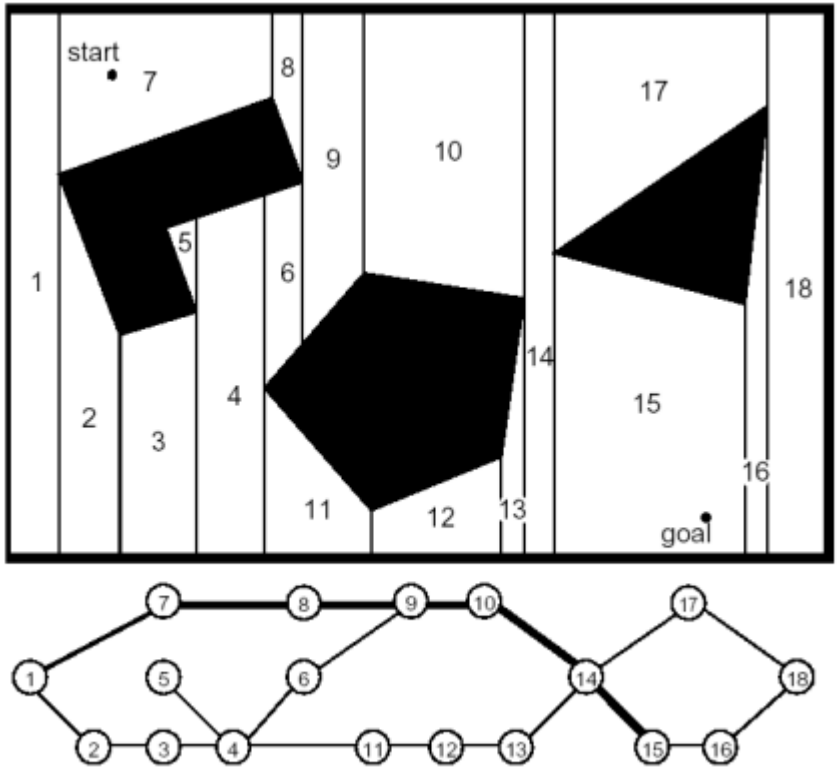
Cell decomposition

Decompose the free space into simple cells and represent the connectivity of the free space by the adjacency graph of these cells

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Cell Decomposition

- Divide the space into simple, connected cells.
- Determine which cells are adjacent and construct a connectivity graph.
- Find cells in which the initial and goal states lie and search for a path in the connectivity graph that joins them.
- From the sequence of cells found with an appropriate search algorithm, compute a path within each cell



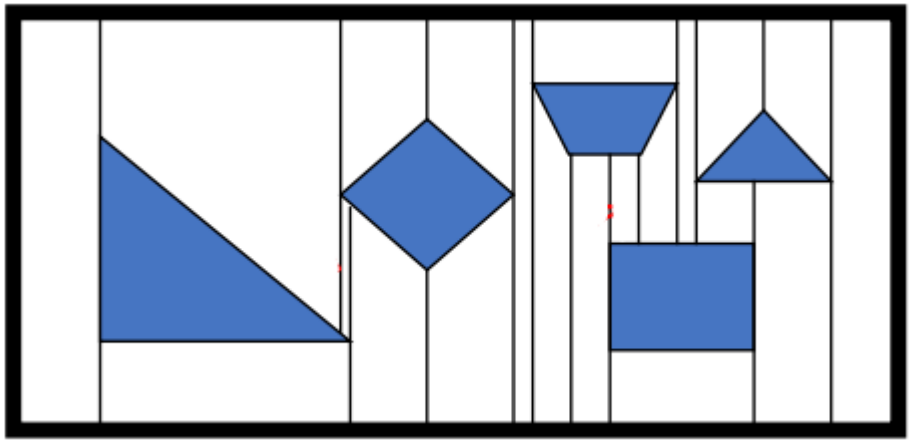
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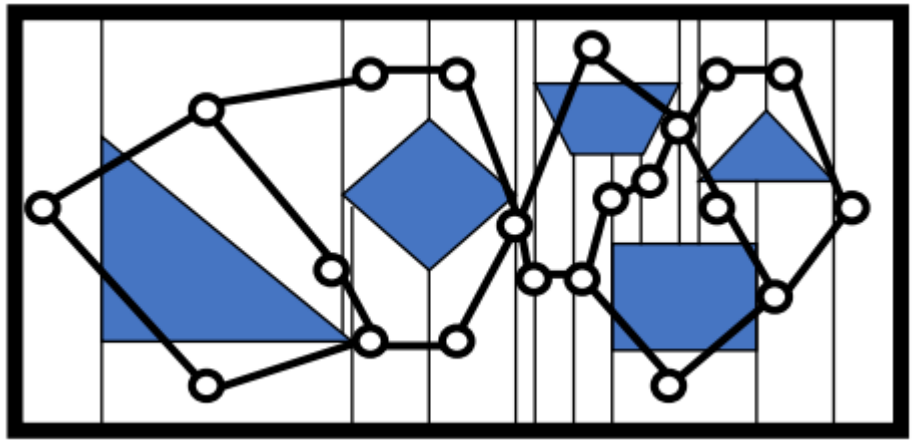
Methods for cell decomposition

- Exact cell decomposition
- Approximate cell decomposition
- Adaptive cell decomposition

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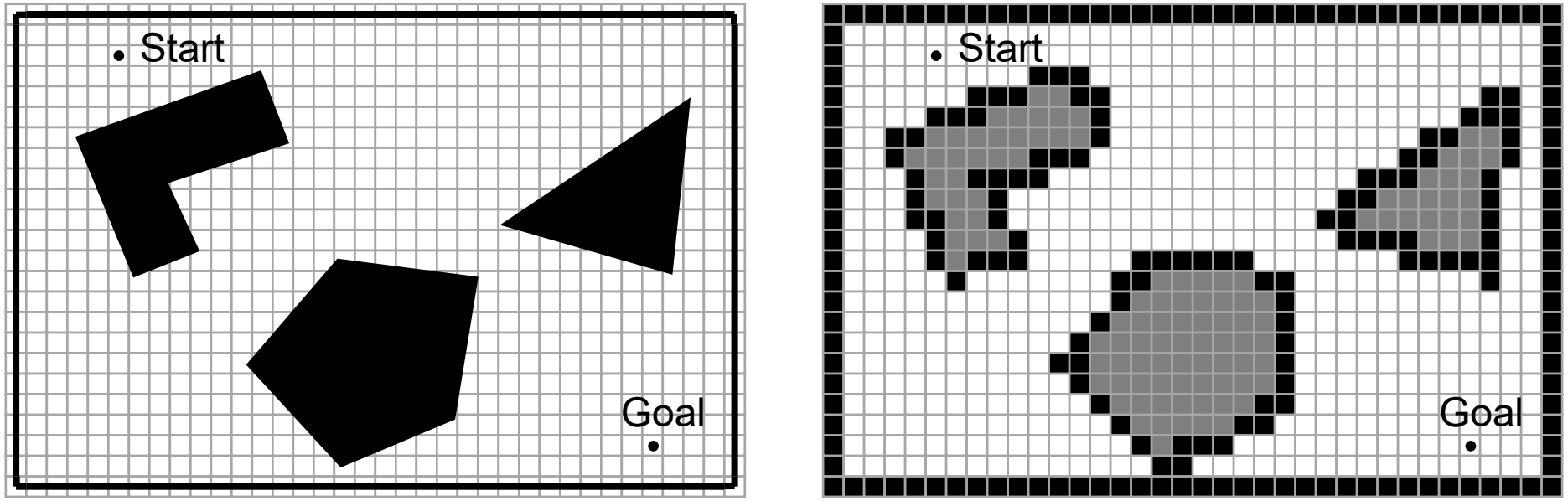
Exact cell decomposition





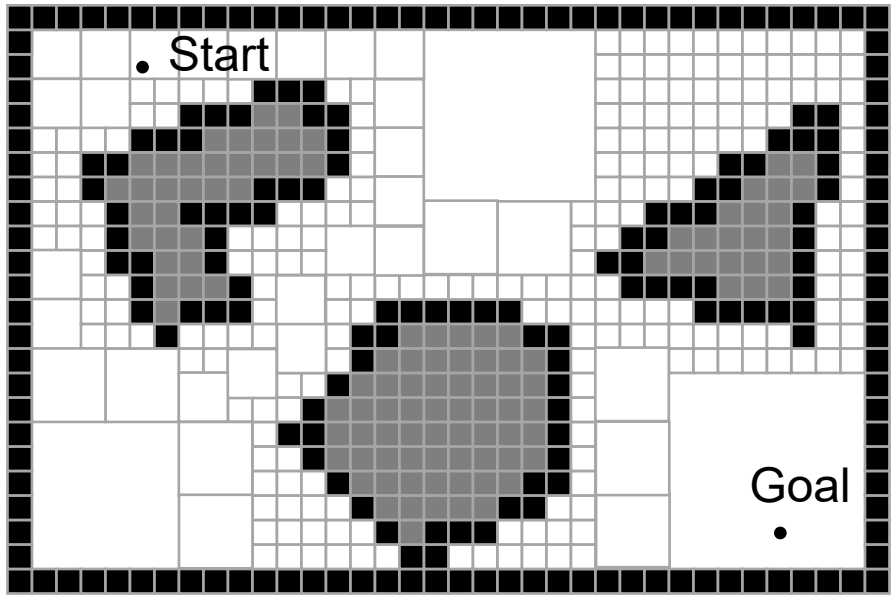
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Approximate cell decomposition



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Adaptive cell decomposition



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Graph search algorithms

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Graph search methods

- Overview
  - Solves a least cost problem between two states on a (directed) graph.
  - Graph structure is a discrete representation.

- Limitations
  - State space is discretised → completeness is at stake.
  - Feasibility of paths is not often encoded.

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Graph search methods

- Deterministic methods
  - Breadth First Search (BFS)
  - Depth First Search (DFS)
  - Dijkstra
  - A*
  - D*

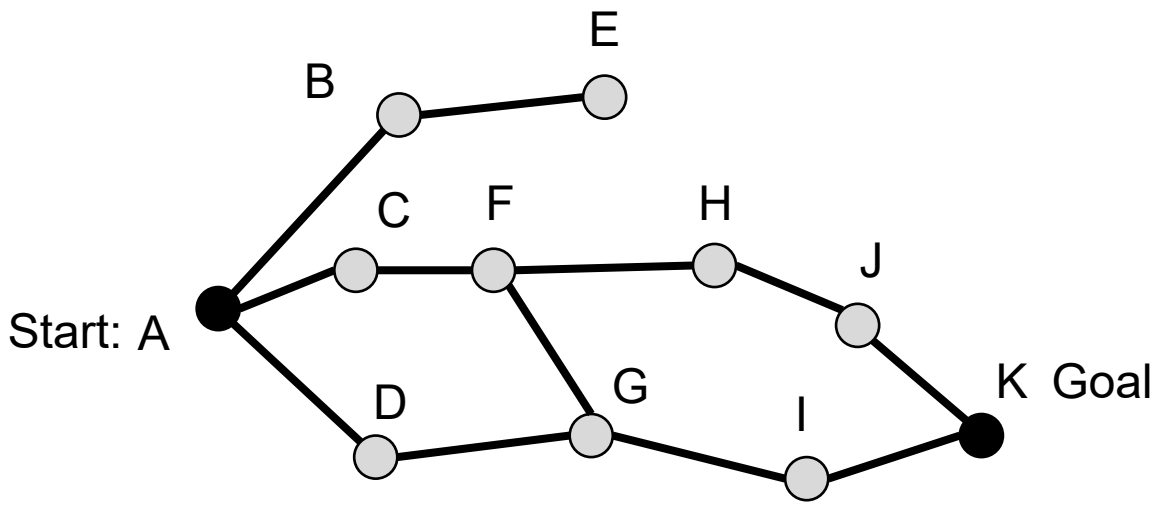
uninformed

informed

- Randomised graph search
  - PRM
  - RRT

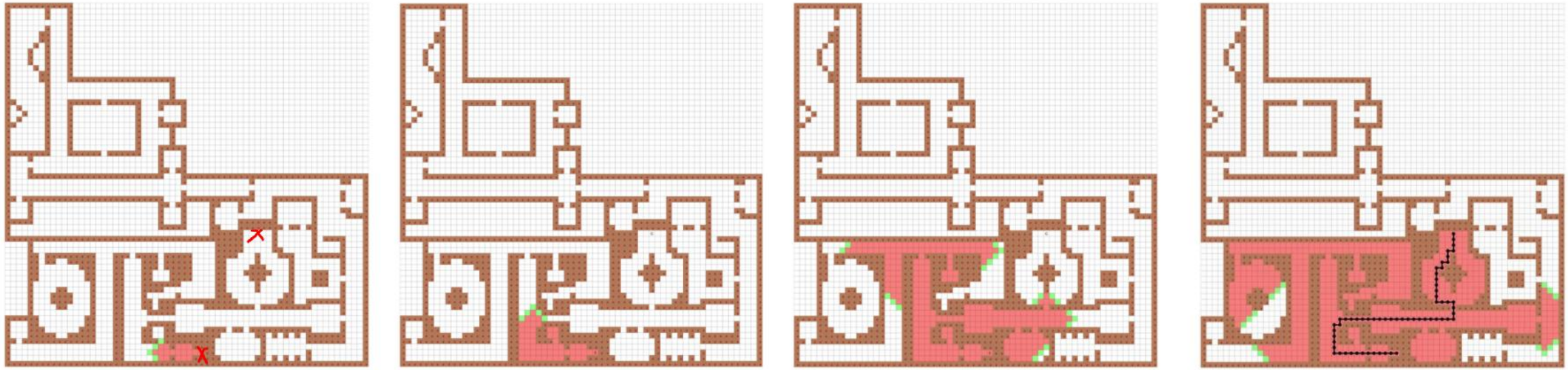
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Breadth first search



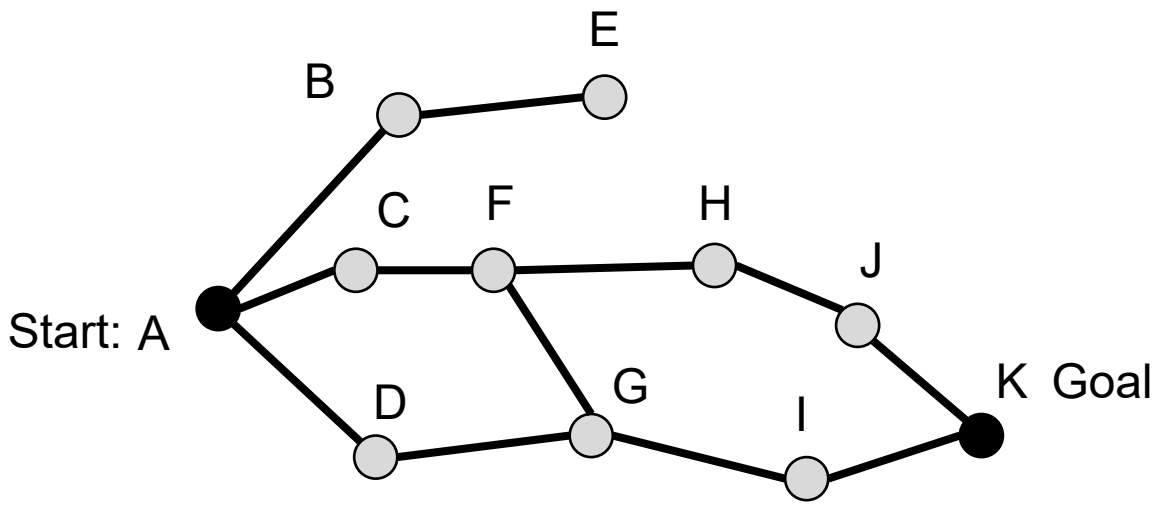
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Example of BFS



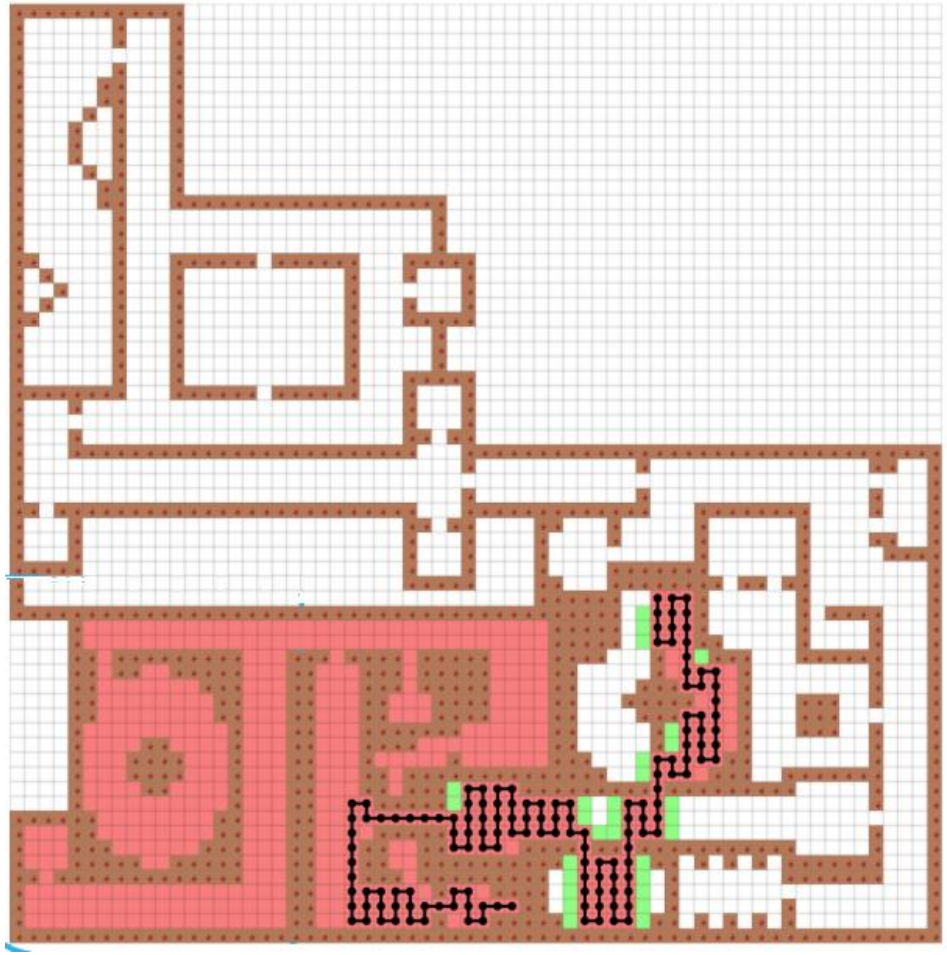
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Depth First Search



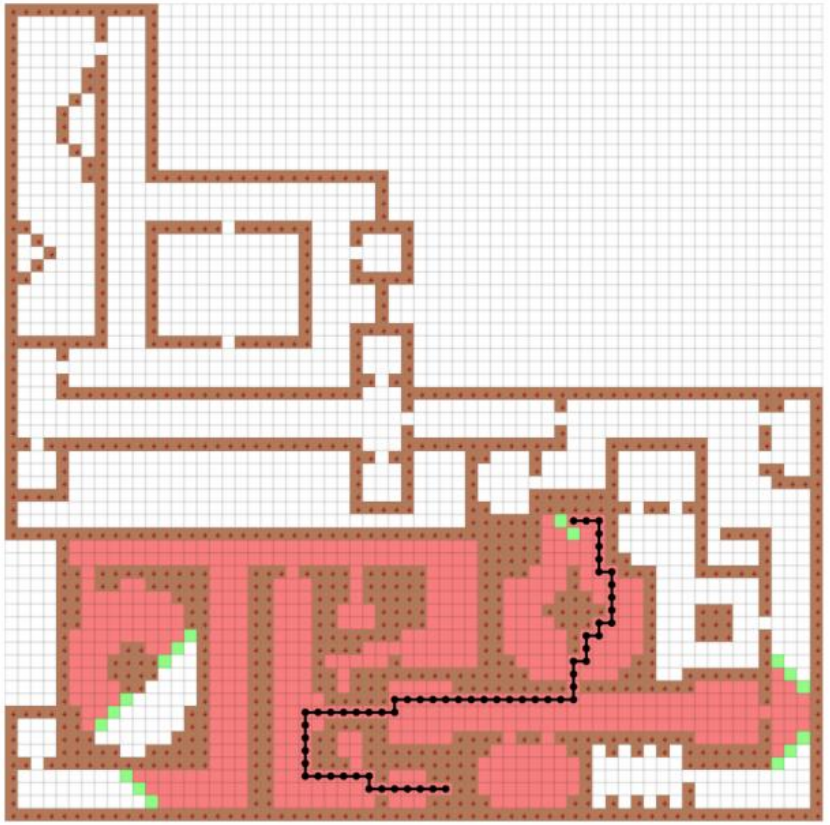
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DFS Example



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Can limit the depth of search



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Weighted graphs

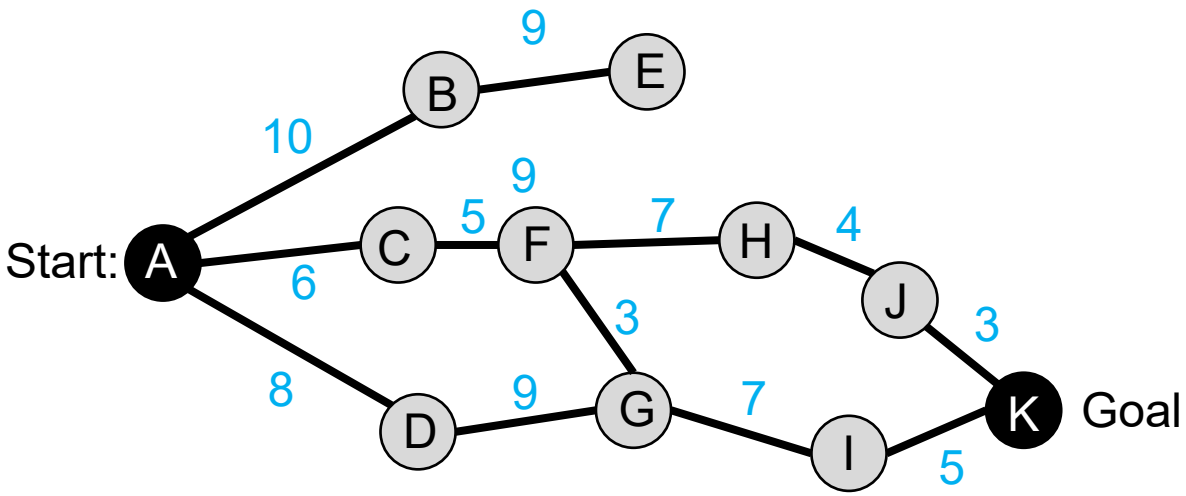
- Each edge has an associated numerical value.
- The weight may represent a cost, distance, etc.
- E.g., in a flight map, the weights may be the distance in miles between places.

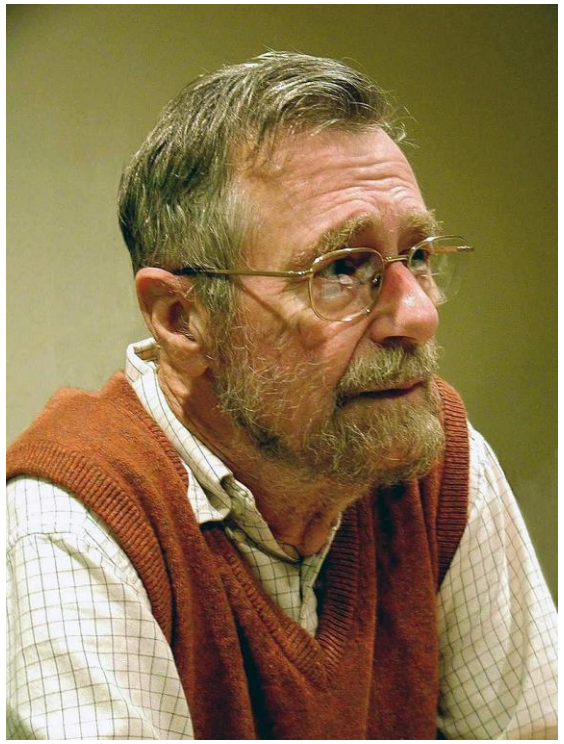
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Dijkstra's algorithm

Edsger Dijkstra in 1959

Search outwards in order of cost from start



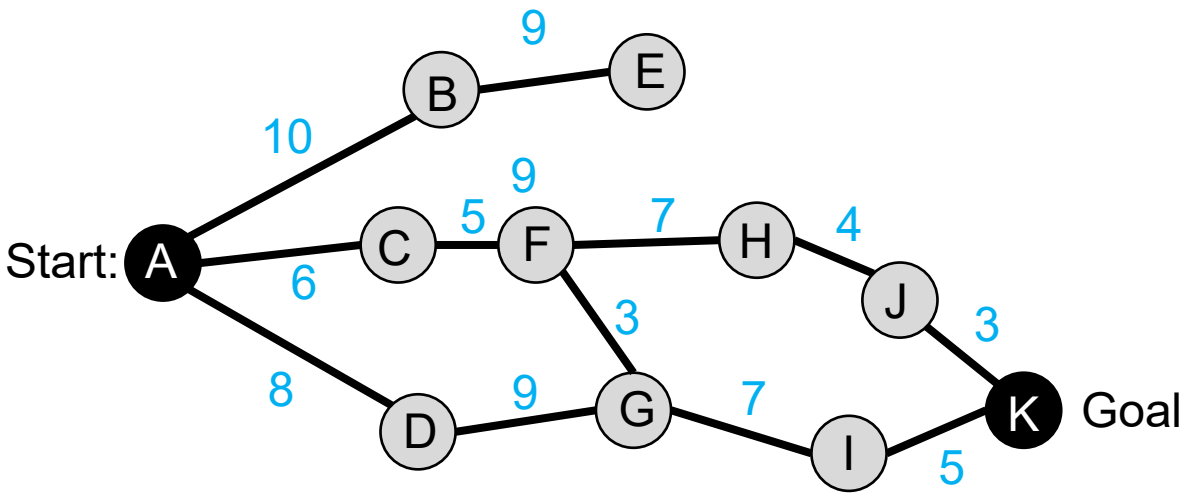


By Hamilton Richards - manuscripts of Edsger W. Dijkstra, University Texas at Austin, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=4204157

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Dijkstra's algorithm

	A	B	C	D	E	F	G	H	I	J	K
Past cost											
Parent node											



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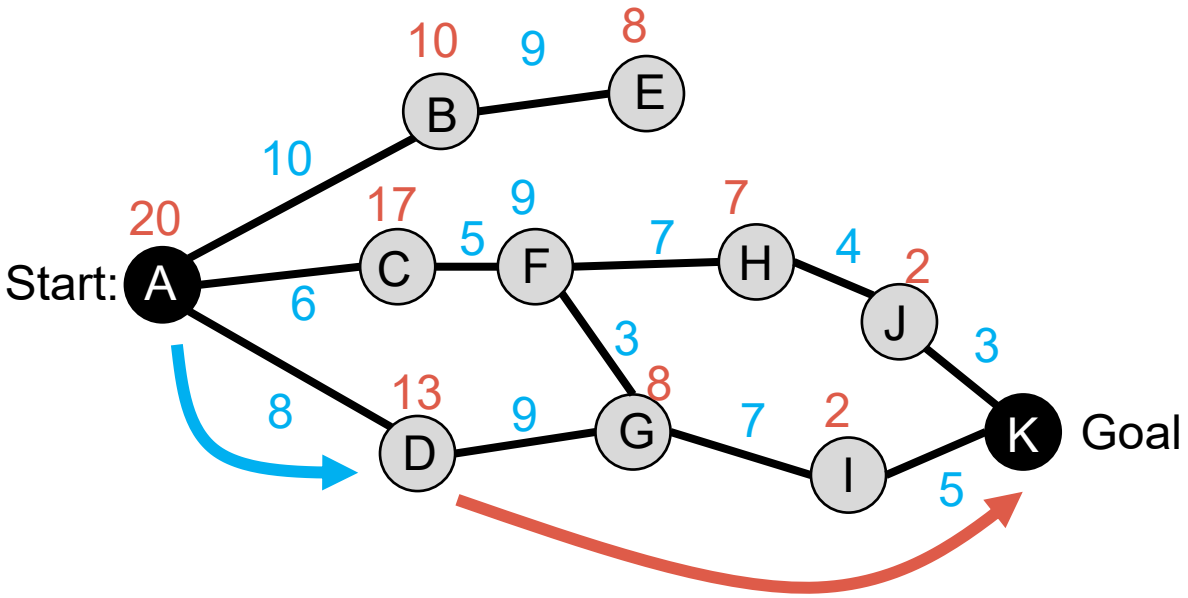
Dijkstra's algorithm

- Doesn't really know the goal exists until it reaches it
- Can we guide the search to expand nodes that are closer to the goal earlier?
- Can we do it without breaking the condition that a node is only accepted with its lowest cost of arrival?

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A* algorithm

Focused search that prioritises searching towards the goal, exploiting some knowledge about it.

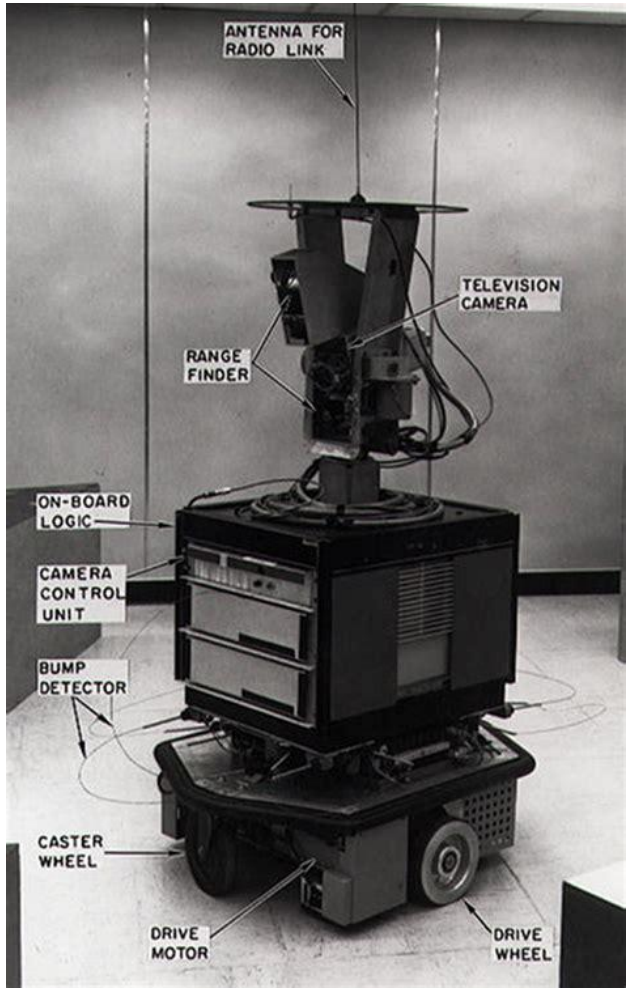


A* priority:

$f(n) = g(n) + h(n)$

Cost to arrive at node n

Optimistic cost from node n to goal



Shakey robot

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A* algorithm

	A	B	C	D	E	F
Past cost						
Optimistic cost to goal						
Estimated total cost						
Parent node						

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Heuristic (Optimistic cost to goal)

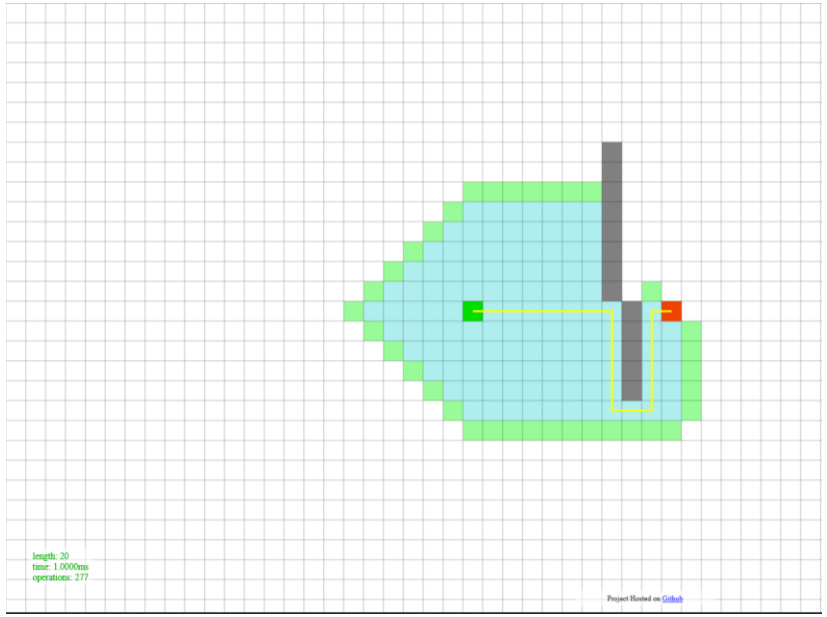
- Needs to be fast to calculate
- Needs to be reasonably close to the real cost
- Needs to be admissible i.e., it shouldn't overestimate the actual cost. $h(n) \leq d(n, goal)$
- Needs to be consistent i.e., for any pair of adjacent nodes n and m, where $d(n, m)$ is the cost of edge between them, $h(n) \leq d(n, m) + h(m)$

Usually defined using some knowledge of the workspace

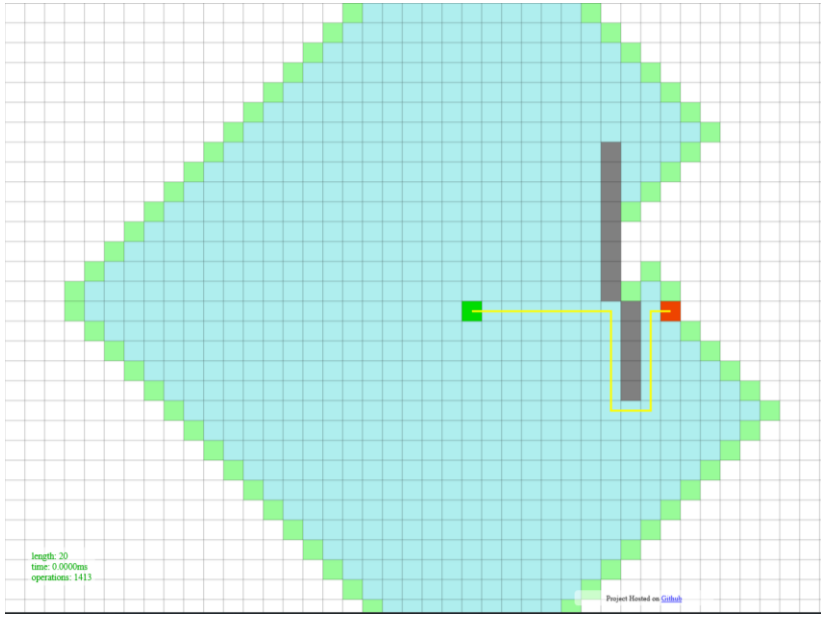
- Euclidian distance
- Manhattan distance
- Zero (Dijkstra's algorithm)

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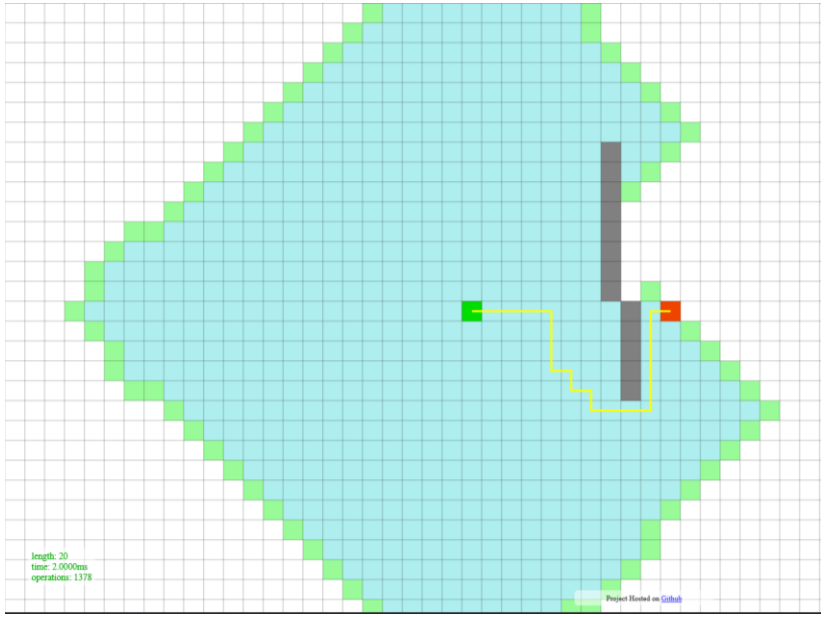
Comparison



A* (277 operations)



BFS (1413 operations)



Dijkstra's (1378 operations)

https://qiao.github.io/PathFinding.js/visual/

Sampling-based planning

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What is wrong with grid-based graphs?

- Can be difficult to select a grid resolution to balance between
  - being sufficiently fine to retain valid solutions (obstacles inflated by grid size)
  - and sufficiently coarse to find a solution in a reasonable time
- Connectivity being defined by the grid shape can be limiting
- Especially challenging in higher dimensional spaces – grids suffer badly from the curse of dimensionality

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Example of the dimensionality problem

- Time complexity of BFS algorithm in a uniform grid is proportional to the sum of the number of edges and vertices in the graph. $O(|V| + |E|)$
- Say we have a 2D grid with 100x100 cells, there are 10^4 nodes. Each cell can have either 4 or 8 neighbours.
- If the grid is 3D with 100x100x100 cells there are 10^6 nodes. A cell can have between 6 and 26 neighbours.
- In a 6D grid of 100 cells per dimension there are 10^{12} nodes! Neighbours?

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Main idea behind sampling-based planning

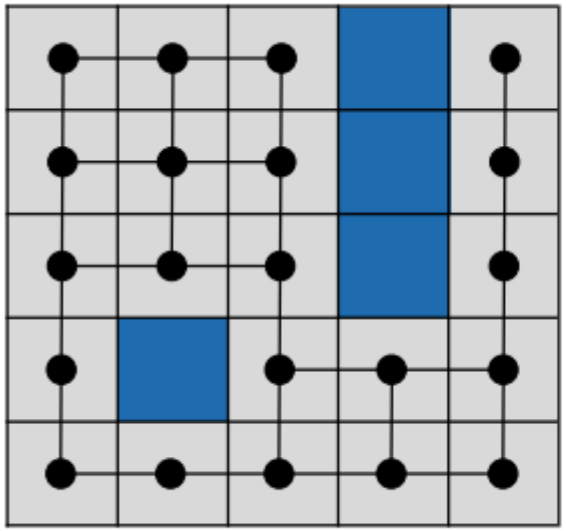
- Instead of dividing the configuration space into uniform (or ordered) cells, randomly pick points in the space and try to connect the samples.
- Often, a large portion of the space is free. If two points are near each other, they can be connected by simple paths like a straight line.

- Abandon the idea of explicitly dividing $\mathcal{C}$ space into $\mathcal{C}_{free}$ and $\mathcal{C}_{obs}$.
- A collision detection algorithms determines if a picked point lies in $\mathcal{C}_{free}$ or not.

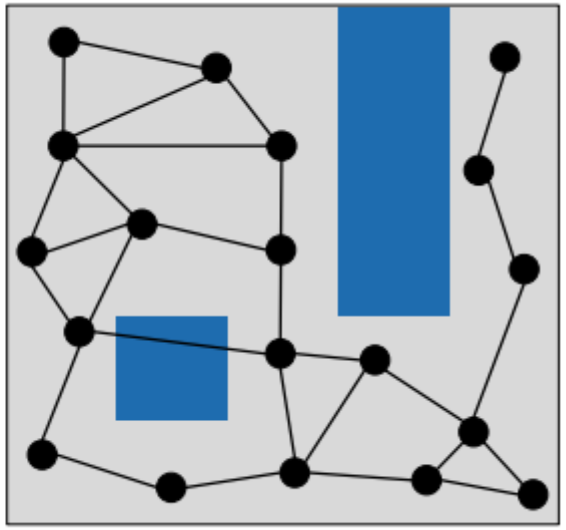
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Random sampling of $\mathcal{C}$ space

Grid-based graph



Sampling-based graph



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Sampling-based planning methods

The two most influential sampling-based motion planning algorithms to date are:

- Probabilistic Roadmaps (1996) - PRM
- Rapidly-exploring random trees (1998) - RRT

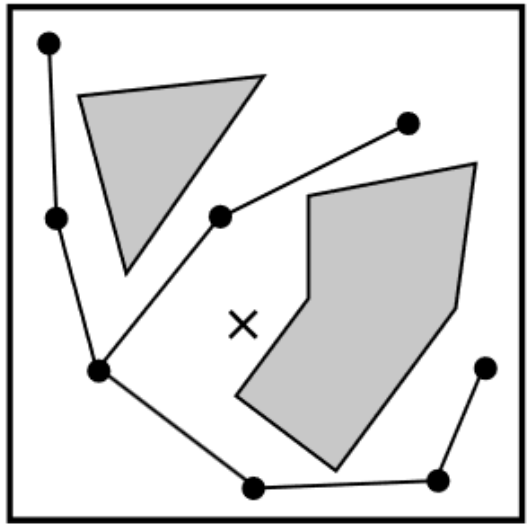
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The general framework

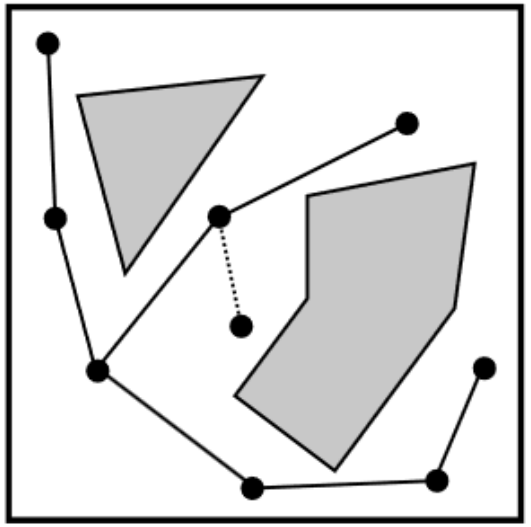
1. Initially the graph G = (V,E) is empty.
2. Then a random point is sampled and added to the set of vertices V.
   Needs a sampling algorithm.
3. For each new vertex v, select a number of nodes from V that are ‘near’ to v.
   Try to connect them to v, using a local planner.
   Needs some function to define ‘near’.
4. If the local path is free of collisions, add the path to the set of edges, E.
   Needs a way to detect collisions.
5. Repeat from step 2 until satisfied.
6. Query for a solution – a connection between start and goal.

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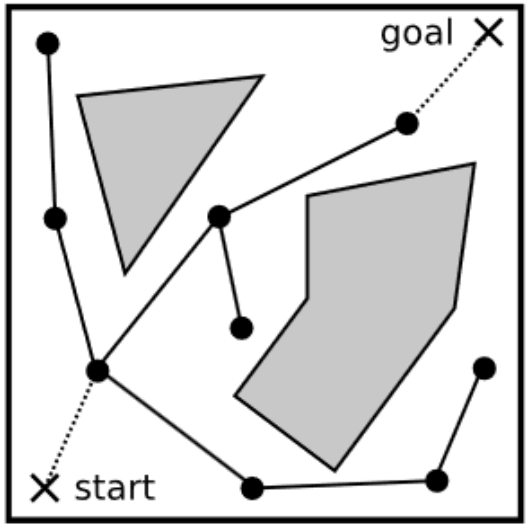
The PRM algorithm



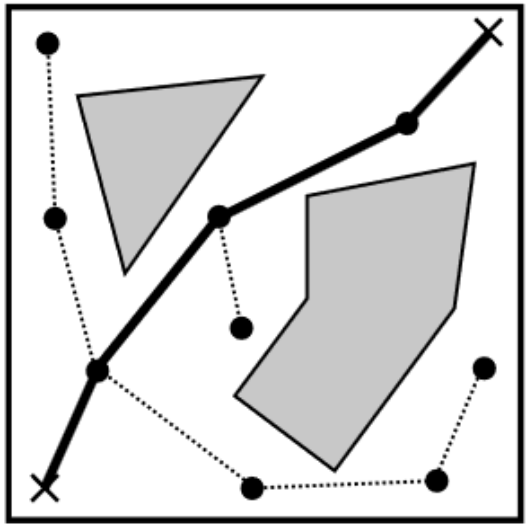
(a) The *learning* phase: a random sample, denoted by ×, is generated



(b) A local planner is used to connect the new sample to nearby roadmap vertices.



(c) The *query* phase: the start and goal configurations are added to the roadmap.



(d) A graph search algorithm is used to connect the start and goal through the roadmap.

From: Short et al. *Recent progress on sampling based dynamic motion planning algorithms. 2016* doi:10.1109/aim.2016.7576950

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The PRM algorithm

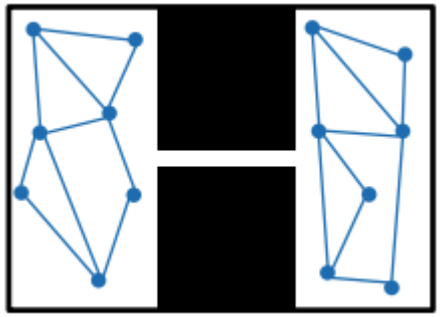
- Build a roadmap by connecting nearby configurations using simple planners to construct a graph of valid path segments.
- Once the roadmap is ready, search the graph. E.g., using A*

Kavraki, Lydia E., et al. "Probabilistic roadmaps for path planning in high-dimensional configuration spaces." IEEE Transactions on Robotics and Automation 12.4 (1996): 566-580.

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PRM - discussion

- Conceptually very simple, and became rapidly popular due to PRM's ability to solve previously challenging problems (high-dimension planning, snake robots)
- Limitations (addressed in subsequent versions of the algorithm)
  - Assumes holonomic motion (i.e., motion is possible in all directions)
  - Can fail to solve a narrow passage problem.
  - Not suited for dynamic environments – roadmap is made first, then searched



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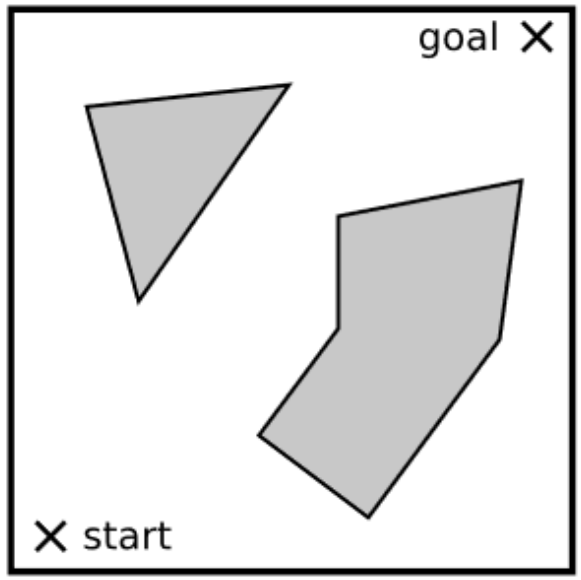
Tree-based planning

- Trees are special types of graphs
- Instead of sampling randomly all over the place, begin at a start configuration, and incrementally grow outwards from there.
- Form a tree of connected, dynamically feasible local paths.
- Terminate when a configuration is reached in a goal region.

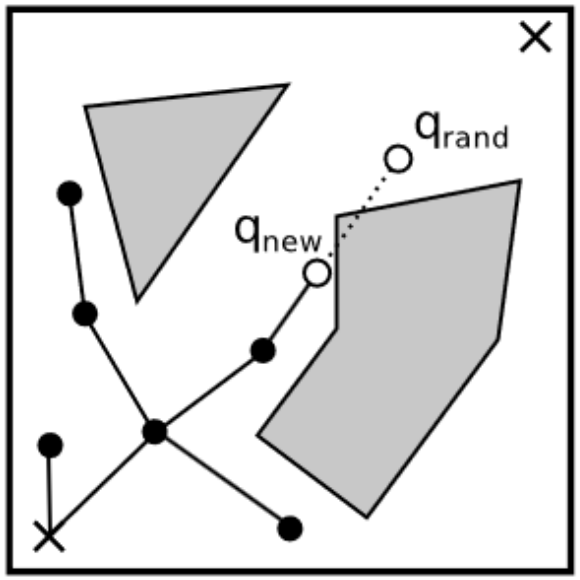
S. M. LaValle. Rapidly-exploring random trees: A new tool for path planning. TR 98- 11, Computer Science Dept., Iowa State Univ., Oct. 1998.

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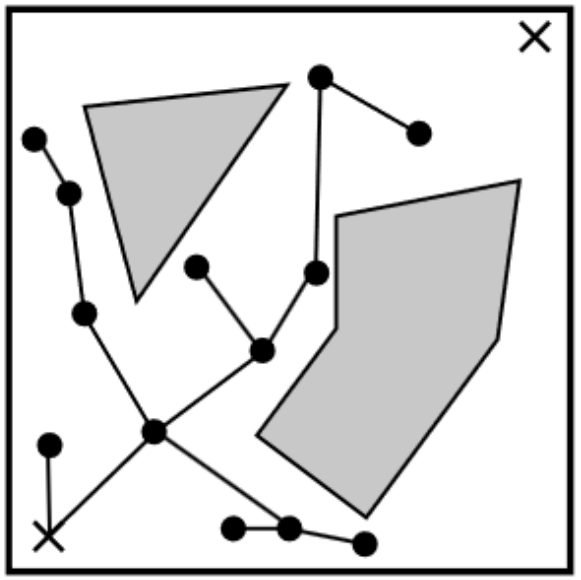
The RRT Algorithm



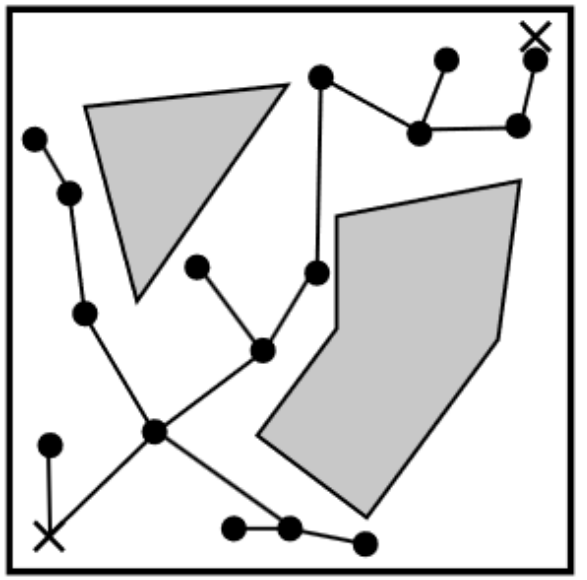
(a) A tree is grown from the start configuration towards the goal.



(b) The planner generates a configuration q_{rand}, and grows from the nearest node towards it to create q_{new}.



(c) The tree rapidly explores the free space.

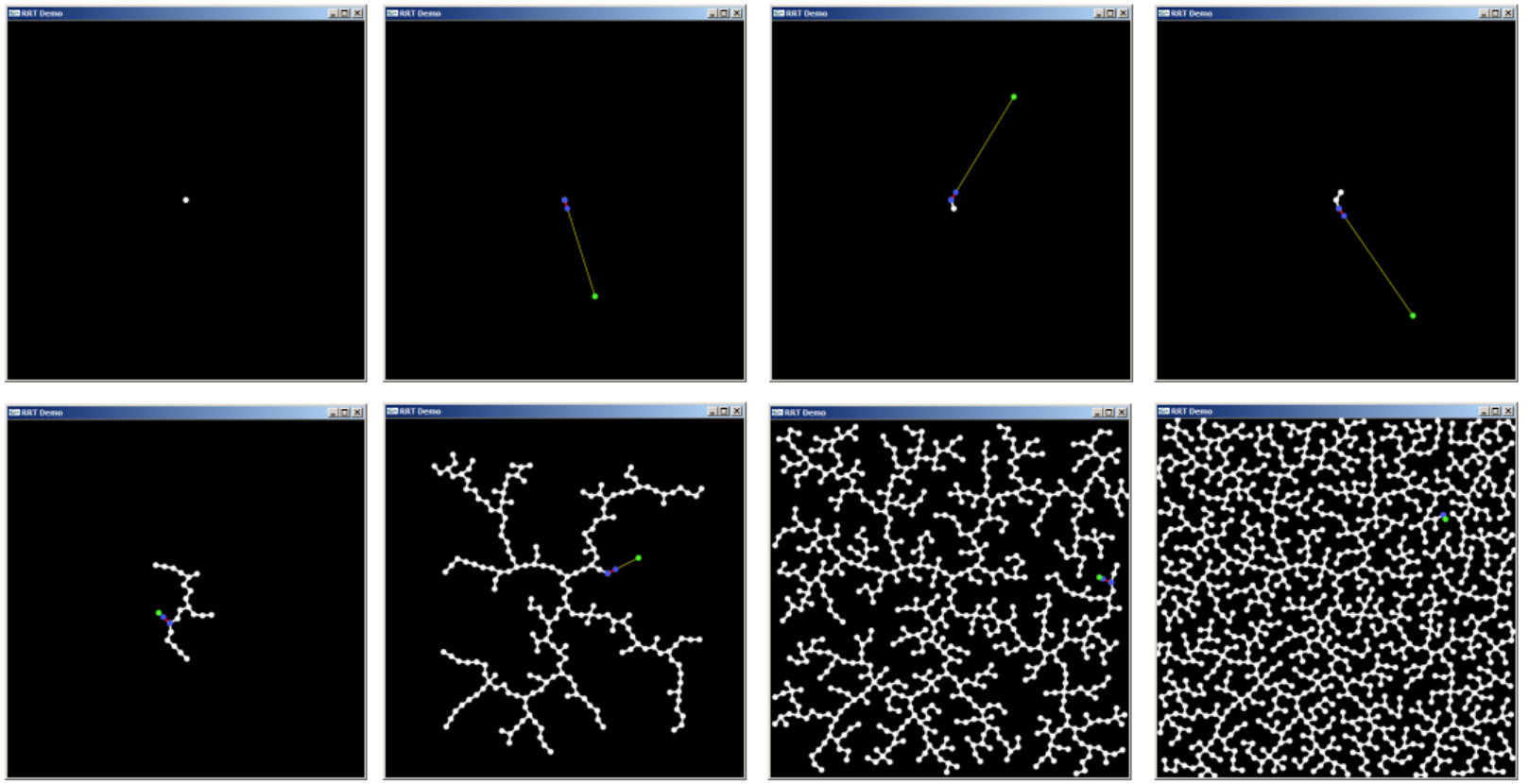


(d) The planner terminates when a node is close to the goal node. Common implementations will connect directly to the goal.

From: Short et al. *Recent progress on sampling based dynamic motion planning algorithms. 2016* doi:10.1109/aim.2016.7576950

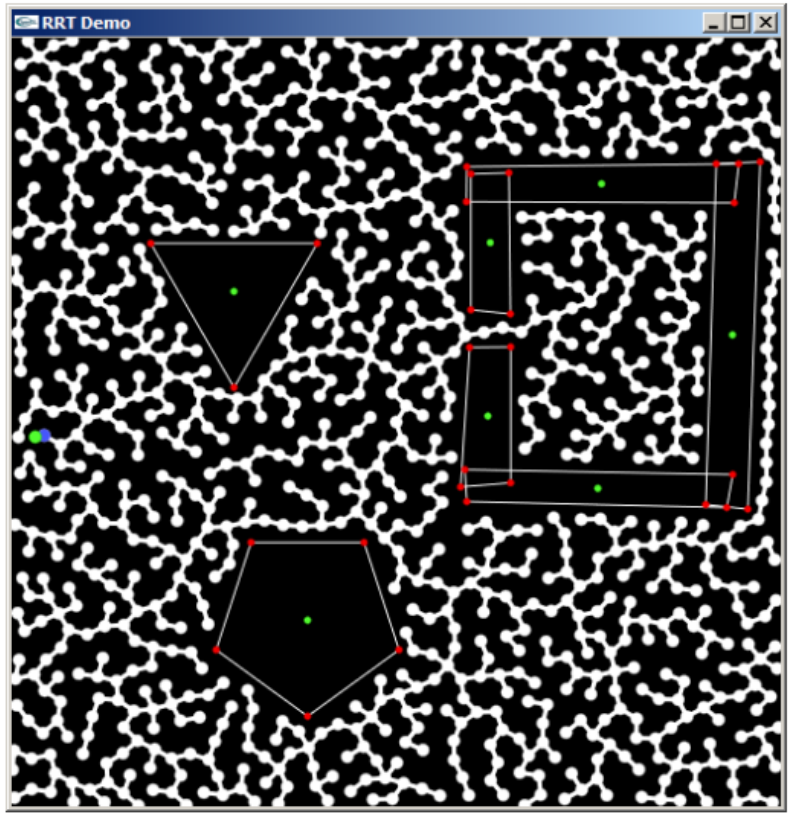
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RRT example



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RRT in the presence of obstacles



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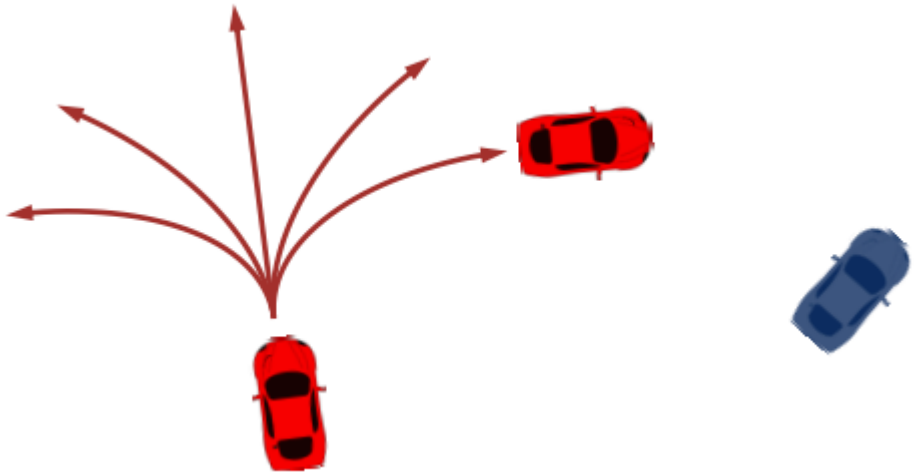
RRTs are good for planning in non-holonomic robots

- We simplified the robot to a point and draw lines in the configuration space.
- This assumes that the robot is free to move in any direction (holonomic), if it doesn’t hit an obstacle.
- However, robots may be non-holonomic.
- How do we solve this problem?

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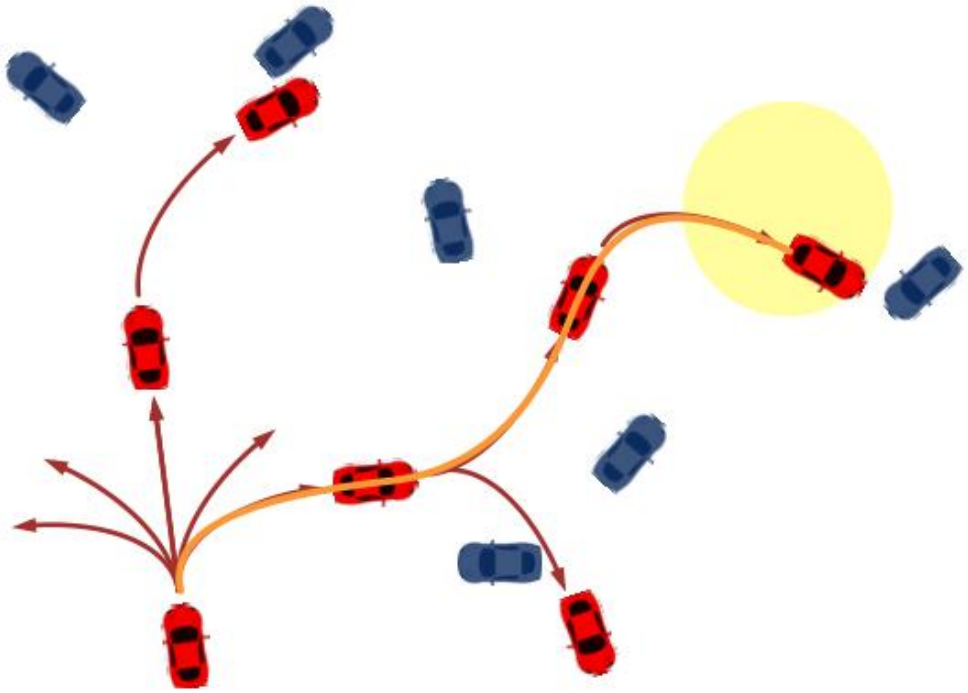
RRTs are good for planning in non-holonomic robots

Instead of trying to join to specific samples, we just sample our controls to generate feasible local paths and choose the configuration that gets closest to the target configuration



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RRT for a car robot



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Completeness of a planner

A path planner is complete if it finds a collision-free path whenever one exists and returns a failure otherwise.

Weaker notions of completeness:

Resolution completeness

- A resolution complete planner discretises the space and returns a path wherever one exists at a chosen resolution

Probabilistic completeness

- A probabilistically complete planner returns a path with high probability if it exists. It may not terminate if no path exists.

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Completeness

- Visibility graphs and Voronoi diagrams are complete.
- BFS – resolution complete.
- DFS – not complete for infinite trees.
- Both PRM and RRT are probabilistically complete – if the number of samples approaches infinity, the probability of samples approaches 1.

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Summary

- We discussed the concept of configuration space
  - To represent the robot and environment in a common space
  - It allows us to think about the robot as point
  - We can then employ the same algorithms to plan in different robots: articulated, car-like, flying, etc.
- We introduced the notion of a graph to model the environment and solve for the planning problem
- For grid-based problems we discussed about
  - Breadth-first search (BFS)
  - Depth-first search (DFS)
  - Dijkstra's Algorithm (for weighted graphs)
  - A*(introduces a heuristic to go faster towards the goal)

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Summary

- We introduced the idea of sampling-based planning approaches
  - More efficient than grid-based
  - We don’t need an explicit representation of the obstacles
- We saw two different approaches
  - Probabilistic roadmaps
  - Rapidly-exploring Random Trees
- Showed how to use RRT with non-holonomic systems

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